

FASE 0 —.

- Definir ordenadores/segmentos IP

Host	IP	Nombre
Kali 1	192.168.56.70	Gastón
Kali 2	192.168.56.75	Álvaro
Kali 3	192.168.56.40	Daniela
Kali 4	192.168.56.17	Dani
Kali 5	192.168.56.22	Rafa

- Documentar alcance y reglas (qué está permitido).
- Crear repositorio con estructura: recon/, pcaps/, logs/, report/.

Este equipo > Documentos > GitHub > Repositorio-Grupo				
	Nombre	Fecha de modificación	Tipo	Tamaño
✦	Ejercicio ficheros	03/10/2025 8:33	Carpeta de archivos	
✦	logs	03/10/2025 8:34	Carpeta de archivos	
✦	pcaps	03/10/2025 8:33	Carpeta de archivos	
✦	recon	03/10/2025 8:33	Carpeta de archivos	
✦	report	03/10/2025 8:34	Carpeta de archivos	

Enlace al documento word del grupo completo:

https://docs.google.com/document/d/1qFGQddWwig2uovvcHGtU73LT-3E_eHnqQp2YDceZ2v8/edit?usp=sharing

FASE 1 — Reconocimiento y mapeo de red

Objetivo: identificar hosts activos, puertos y servicios.

Herramientas y comandos (ejemplos):

- Descubrimiento de hosts:
 - **arp-scan (si está permitido:**
 - `sudo arp-scan -l`

```
(kali@kali)-[~]
$ sudo arp-scan -l
Interface: eth0, type: EN10MB, MAC: 08:00:27:75:a0:90, IPv4: 192.168.56.40
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.56.22 08:00:27:86:53:97 (Unknown)
192.168.56.17 08:00:27:99:2c:ef (Unknown)
192.168.56.23 08:00:27:92:38:45 (Unknown)
192.168.56.70 08:00:27:7e:fa:03 (Unknown)
192.168.56.75 08:00:27:5f:6d:09 (Unknown)
192.168.56.80 08:00:27:61:0f:54 (Unknown)
192.168.56.81 08:00:27:45:bc:2b (Unknown)
192.168.56.82 08:00:27:59:48:c3 (Unknown)

8 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.814 seconds (141.12 hosts/sec). 8 responded
```

- **netdiscover:**
 - `sudo netdiscover -r 192.168.56.0/24`

```
Currently scanning: Finished! | Screen View: Unique Hosts
770 Captured ARP Req/Rep packets, from 21 hosts. Total size: 46200
```

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.0.115	18:60:24:e9:3b:2a	4	240	Hewlett Packard
192.168.0.123	34:17:eb:c4:6c:81	51	3060	Dell Inc.
192.168.56.22	08:00:27:86:53:97	4	240	PCS Systemtechnik GmbH
192.168.56.23	08:00:27:92:38:45	5	300	PCS Systemtechnik GmbH
192.168.56.70	08:00:27:7e:fa:03	3	180	PCS Systemtechnik GmbH
192.168.56.75	08:00:27:5f:6d:09	1	60	PCS Systemtechnik GmbH
192.168.56.80	08:00:27:61:0f:54	70	4200	PCS Systemtechnik GmbH
192.168.56.81	08:00:27:45:bc:2b	1	60	PCS Systemtechnik GmbH
192.168.56.82	08:00:27:59:48:c3	15	900	PCS Systemtechnik GmbH
192.168.0.67	18:60:24:e9:3a:50	30	1800	Hewlett Packard
192.168.0.56	00:e0:4c:68:0f:5f	12	720	REALTEK SEMICONDUCTOR CORP.
192.168.0.65	e8:39:35:57:9e:e9	30	1800	Hewlett Packard
192.168.0.85	08:ed:ed:b6:33:cc	2	120	Zhejiang Dahua Technology Co., Ltd
192.168.0.73	dc:4a:3e:7c:25:3e	13	780	Hewlett Packard
192.168.0.78	dc:4a:3e:8d:c1:b5	10	600	Hewlett Packard
192.168.0.108	68:34:21:82:4c:7c	1	60	Intel Corporate
192.168.56.17	08:00:27:99:2c:ef	1	60	PCS Systemtechnik GmbH
192.168.0.55	08:ed:ed:b5:aa:e6	1	60	Zhejiang Dahua Technology Co., Ltd
192.168.0.90	a4:97:33:4e:9f:c4	510	30600	ASKEY COMPUTER CORP
0.0.0.0	08:00:27:86:53:97	3	180	PCS Systemtechnik GmbH
0.0.0.0	08:00:27:92:38:45	3	180	PCS Systemtechnik GmbH

- Escaneo de puertos con nmap:
 - **Scan rápido de puertos y servicios:**
 - `nmap -sS -sV -O -Pn 192.168.56.0/24`

```
(kali㉿kali)-[~]  
$ nmap -sS -sV -O -Pn 192.168.56.0/24  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-03 13:35 CEST  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.  
Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.22  
Host is up (0.00078s latency).  
All 1000 scanned ports on 192.168.56.22 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
MAC Address: 08:00:27:86:53:97 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 1 hop
```

- Host de Gastón:

```
Nmap scan report for 192.168.56.70  
Host is up (0.00078s latency).  
All 1000 scanned ports on 192.168.56.70 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
MAC Address: 08:00:27:7E:FA:03 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 1 hop
```

- Host de Álvaro:

```
Nmap scan report for 192.168.56.75  
Host is up (0.0014s latency).  
All 1000 scanned ports on 192.168.56.75 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
MAC Address: 08:00:27:5F:6D:09 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 1 hop
```

- Host mío:

```
Nmap scan report for 192.168.56.40  
Host is up (0.000063s latency).  
All 1000 scanned ports on 192.168.56.40 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 0 hops
```

- Escaneo de puertos específicos:
 - `nmap -p 22,80,443,3128 192.168.56.101`

```
(kali@kali)-[~]
└─$ nmap -p 22,80,443,3128 192.168.56.70
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-03 13:59 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.70
Host is up (0.00094s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
80/tcp    closed http
443/tcp   closed https
3128/tcp  closed squid-http
MAC Address: 08:00:27:7E:FA:03 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.17 seconds

(kali@kali)-[~]
└─$ nmap -p 22,80,443,3128 192.168.56.75
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-03 13:59 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.75
Host is up (0.00087s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
80/tcp    closed http
443/tcp   closed https
3128/tcp  closed squid-http
MAC Address: 08:00:27:5F:6D:09 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.17 seconds
```

```
└─$ nmap -p 22,80,443,3128 192.168.56.17
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-03 13:59 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.17
Host is up (0.0025s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
80/tcp    closed http
443/tcp   closed https
3128/tcp  closed squid-http
MAC Address: 08:00:27:6D:45:BB (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.18 seconds

(kali@kali)-[~]
└─$ nmap -p 22,80,443,3128 192.168.56.22
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-03 13:59 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.22
Host is up (0.00092s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
80/tcp    closed http
443/tcp   closed https
3128/tcp  closed squid-http
MAC Address: 08:00:27:86:53:97 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds

(kali@kali)-[~]
└─$ nmap -p 22,80,443,3128 192.168.56.40
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-03 14:01 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled.
Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.40
Host is up (0.00022s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
80/tcp    closed http
443/tcp   closed https
3128/tcp  closed squid-http

Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
```

- Resultado esperado: lista de hosts, puertos abiertos, servicios y versiones.

FASE 2 — Análisis de tráfico y paquetes

Objetivo: capturar tráfico relevante y analizar patrones/anomalías.

Herramientas y comandos:

- **Captura de paquetes:**
 - **tcpdump para capturar en interfaz:**
 - `sudo tcpdump -i eth0 -w /home/kali/pcaps/scan_capture.pcap`

```
(kali㉿kali)-[~]
$ sudo tcpdump -i eth0 -w /home/kali/pcaps/scan_capture.pcap

tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C788 packets captured
790 packets received by filter
0 packets dropped by kernel
```

- WireShark
 - Filtro tráfico relevante

1990	97.315159956	192.168.56.40	192.168.56.22	TCP
1985	97.314979258	192.168.56.40	192.168.56.22	TCP
4018	98.932888995	192.168.56.22	192.168.56.40	TCP
4016	98.907866967	192.168.56.22	192.168.56.40	TCP
4013	98.882583442	192.168.56.22	192.168.56.40	TCP

- **Monitoreo en vivo:**

Dirección			
Dirección	Máscara de red	Puerta de enlace	
192.168.56.40	24	192.168.56.1	Añadir
			Eliminar

```
valid_ltl forever preferred_ltl forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:75:a0:90 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.51/23 brd 192.168.1.255 scope global dynamic noprefixroute eth0
```

Para la mayoría de ejercicios se ha utilizado la red privada con la ip 192.168.56.40, pero al esta no tener acceso a internet, en algunos otros se ha cambiado por la ip 10.0.2.15

- iftop (consumo por conx):

- sudo iftop -i eth0

Session	Acciones	Editar	Vista	Ayuda					
					1,91Mb	3,81Mb	5,72Mb	7,63Mb	9,54Mb
10.0.2.15					⇒	mad41s10-in-f8.1e100.net		0b	0b
					≤			0b	0b
10.0.2.15					⇒	172.67.75.39		0b	0b
					≤			0b	0b
10.0.2.15					⇒	104.16.174.226		0b	0b
					≤			0b	0b
10.0.2.15					⇒	203.137.36.34.bc.googleusercontent		0b	0b
					≤			0b	0b
10.0.2.15					⇒	mad07s10-in-f14.1e100.net		0b	0b
					≤			0b	0b
10.0.2.15					⇒	104.16.79.73		0b	0b
					≤			0b	0b
10.0.2.15					⇒	216.72.190.35.bc.googleusercontent		0b	0b
					≤			0b	0b
10.0.2.15					⇒	185.43.181.145		0b	0b
					≤			0b	0b
10.0.2.15					⇒	254.red-80-58-61.staticip.rima-		0b	0b
					≤			0b	0b
10.0.2.15					⇒	mad07s25-in-f3.1e100.net		0b	0b
					≤			0b	0b
10.0.2.15					⇒	5.255.145.233		0b	0b
					≤			0b	0b
10.0.2.15					⇒	191.144.160.34.bc.googleusercontent		0b	0b
					≤			0b	0b
10.0.2.15					⇒	104.18.38.233		0b	0b
					≤			0b	0b
10.0.2.15					⇒	250.red-80-58-61.staticip.rima-		0b	0b
					≤			0b	0b
TX:					cum:	44,3KB	peak:	131Kb	rates:
RX:						320KB		1,14Mb	
TOTAL:						364KB		1,27Mb	
								0b	0b
								0b	0b
								0b	0b
								8,83Kb	
								63,9Kb	
								72,7Kb	

- nethogs (por proceso)

NetHogs version 0.8.8					
PID	USER	PROGRAM	DEV	SENT	RECEIVED
17564	kali	/usr/lib/firefox-esr/firefox-esr	eth0	3.885	9.019 kB/s
?	root	10.0.2.15:45524-34.120.158.37:443		0.000	0.000 kB/s
?	root	unknown TCP		0.000	0.000 kB/s
TOTAL				3.885	9.019 kB/s

- **Análisis de .pcap:**
 - **Abrir pcap con Wireshark (GUI) o tshark:**
 - `tshark -r scan_capture.pcap -q -z io,phs`

```
(kali㉿kali)-[~]  
$ tshark -r /home/kali/pcaps/scan_capture.pcap -q -z io,phs
```

Protocol Hierarchy Statistics
Filter:

eth	frames:788 bytes:61081
ip	frames:152 bytes:19345
icmp	frames:39 bytes:3822
udp	frames:104 bytes:14983
data	frames:78 bytes:9701
mdns	frames:12 bytes:2522
ssdp	frames:3 bytes:429
nbdgm	frames:7 bytes:1707
smb	frames:7 bytes:1707
mailslot	frames:7 bytes:1707
browser	frames:7 bytes:1707
dhcp	frames:1 bytes:348
nbns	frames:3 bytes:276
igmp	frames:9 bytes:540
arp	frames:587 bytes:35184
ipv6	frames:47 bytes:6068
udp	frames:19 bytes:3616
mdns	frames:12 bytes:2762
dhcpv6	frames:7 bytes:854
icmpv6	frames:28 bytes:2452
lldp	frames:2 bytes:484

FASE 3 — Enumeración de servicios y pruebas seguras

Objetivo: verificar configuraciones visibles (vulnerabilidades de servicio *sin explotar*).

Herramientas:

- nmap scripts (--script), ss, curl, nikto (solo pruebas no destructivas).

```
(kali㉿kali)-[~]  
$ sudo systemctl start apache2  
  
(kali㉿kali)-[~]  
$ sudo systemctl enable apache2  
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/s  
ystemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2  
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' → '/usr/l  
ib/systemd/system/apache2.service'.
```

Ejemplos:

- nmap -sV --script=vuln 192.168.56.101

```
(kali㉿kali)-[~]  
$ nmap -sV --script=vuln 192.168.56.17  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-07 09:51 CEST  
Nmap scan report for 192.168.56.17  
Host is up (0.00067s latency).  
All 1000 scanned ports on 192.168.56.17 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
MAC Address: 08:00:27:6D:45:BB (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
  
Service detection performed. Please report any incorrect results at https://nmap.  
bmit/ .  
Nmap done: 1 IP address (1 host up) scanned in 23.70 seconds
```

- curl -I http://192.168.56.101:80

```
(kali㉿kali)-[~]  
$ curl -I http://192.168.56.17:80  
HTTP/1.1 200 OK  
Date: Tue, 07 Oct 2025 07:58:32 GMT  
Server: Apache/2.4.65 (Debian)  
Last-Modified: Thu, 02 Oct 2025 12:12:41 GMT  
ETag: "29cf-6402be6e1abdd"  
Accept-Ranges: bytes  
Content-Length: 10703  
Vary: Accept-Encoding  
Content-Type: text/html
```

- nikto -host <http://192.168.56.101:80>


```

(kali㉿kali)-[~]
$ nikto -host http://192.168.56.17:80
- Nikto v2.5.0

+ Target IP:      192.168.56.17
+ Target Hostname: 192.168.56.17
+ Target Port:    80
+ Start Time:     2025-10-07 10:01:17 (GMT2)

+ Server: Apache/2.4.65 (Debian)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ /: Server may leak inodes via ETags, header found with file /, inode: 29cf, size: 6402be6elabdd, mtime: gzip. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418
+ OPTIONS: Allowed HTTP Methods: GET, POST, OPTIONS, HEAD .
+ 8102 requests: 0 error(s) and 4 item(s) reported on remote host
+ End Time:      2025-10-07 10:01:43 (GMT2) (26 seconds)

+ 1 host(s) tested

```

Nota: no realizar explotación sin autorización.

FASE 4 — Análisis de logs y forense básico

Objetivo: revisar logs del sistema/servicios para identificar actividad sospechosa.

Ubicaciones y herramientas:

- Logs de sistema: /var/log/syslog, /var/log/auth.log.

```

(kali㉿kali)-[~]
$ cd /var/log

(kali㉿kali)-[/var/log]
$ ls
alternatives.log  gvm          nginx         samba
apache2          inetsim      notus-scanner speech-dispatcher
apt              installer    openvpn       squid
boot.log         journal     postgresql   stunnel4
boot.log.1       lastlog     private      sysstat
btmtp            lightdm      README       wtmp
dpkg.log         macchanger.log redis        Xorg.0.log
fontconfig.log   mosquito    runit        Xorg.0.log.old

```

- Logs de aplicaciones: Squid /var/log/squid/access.log, web /var/log/apache2/access.log.

```
(kali㉿kali)-[~]  
$ squid /var/log/squid/access.log  
  
2025/10/07 10:14:12| Processing Configuration File: /etc/squid/squid.conf (depth 0)  
2025/10/07 10:14:12| Processing Configuration File: /etc/squid/conf.d/debian.conf (depth 1)  
2025/10/07 10:14:12| FATAL: Squid is already running: Found fresh instance PID file (/run/squid.pid) with PID 42577  
exception location: Instance.cc(122) ThrowIfAlreadyRunningWith
```

Herramientas: grep, ausearch(auditd), journalctl, ELK/Graylog (si disponible).

Comandos útiles:

- `sudo tail -n 200 /var/log/auth.log`
- `sudo grep "Failed password" /var/log/auth.log | wc -l`
- `sudo journalctl -u ssh`

Kali (basado en Debian, pero más minimalista) usa systemd-journald para guardar los registros en binario dentro de /var/log/journal/

En lugar de leer un archivo, se usa journalctl:

```
(kali㉿kali)-[~]  
$ sudo journalctl -u ssh  
  
-- No entries --
```

para ver todas las entradas relacionadas con autenticación:

```
(kali㉿kali)-[~]  
$ sudo journalctl | grep sshd  
  
oct 02 14:30:32 kali systemd[1]: Listening on sshd-unix-local.socket -  
OpenSSH Server Socket (systemd-ssh-generator, AF_UNIX Local).  
oct 02 14:30:32 kali systemd[1]: Listening on sshd-vsock.socket - OpenS  
SH Server Socket (systemd-ssh-generator, AF_VSOCK).  
oct 03 14:45:48 kali systemd[1]: Listening on sshd-unix-local.socket -  
OpenSSH Server Socket (systemd-ssh-generator, AF_UNIX Local).  
oct 03 14:45:48 kali systemd[1]: Listening on sshd-vsock.socket - OpenS  
SH Server Socket (systemd-ssh-generator, AF_VSOCK).  
oct 07 08:56:47 kali systemd[1]: Listening on sshd-unix-local.socket -  
OpenSSH Server Socket (systemd-ssh-generator, AF_UNIX Local).  
oct 07 08:56:47 kali systemd[1]: Listening on sshd-vsock.socket - OpenS  
SH Server Socket (systemd-ssh-generator, AF_VSOCK).
```

y para ver intentos de inicio de sesión locales:

```
(kali㉿kali)-[~]  
$ sudo journalctl _COMM=login  
sudo journalctl _COMM=sudo  
  
-- No entries --  
oct 02 14:47:53 kali sudo[2566]:      kali : TTY=pts/0 ; PWD=/home/kali>  
oct 02 14:47:53 kali sudo[2566]: pam_unix(sudo:session): session opene>  
oct 02 14:47:55 kali sudo[2566]: pam_unix(sudo:session): session close>  
oct 02 14:52:15 kali sudo[4766]:      kali : TTY=pts/0 ; PWD=/home/kali>  
oct 02 14:52:15 kali sudo[4766]: pam_unix(sudo:session): session opene>  
oct 02 14:52:26 kali sudo[4766]: pam_unix(sudo:session): session close>  
oct 03 11:04:20 kali sudo[86980]:      kali : TTY=pts/0 ; PWD=/home/ka>  
oct 03 11:04:20 kali sudo[86980]: pam_unix(sudo:session): session open>  
oct 03 11:04:20 kali sudo[86980]: pam_unix(sudo:session): session clos>  
oct 03 13:16:45 kali sudo[117275]:      kali : TTY=pts/0 ; PWD=/home/ka>  
oct 03 13:16:45 kali sudo[117275]: pam_unix(sudo:session): session ope>  
oct 03 13:16:47 kali sudo[117275]: pam_unix(sudo:session): session clo>  
oct 03 13:21:51 kali sudo[119761]:      kali : TTY=pts/0 ; PWD=/home/ka>  
oct 03 13:21:51 kali sudo[119761]: pam_unix(sudo:session): session ope>  
oct 03 13:21:51 kali sudo[119761]: pam_unix(sudo:session): session clo>  
oct 03 13:22:19 kali sudo[120004]:      kali : TTY=pts/0 ; PWD=/home/ka>  
oct 03 13:22:19 kali sudo[120004]: pam_unix(sudo:session): session ope>  
oct 03 13:22:21 kali sudo[120004]: pam_unix(sudo:session): session clo>  
oct 03 13:22:30 kali sudo[120100]:      kali : TTY=pts/0 ; PWD=/home/ka>  
oct 03 13:22:30 kali sudo[120100]: pam_unix(sudo:session): session ope>  
oct 03 13:22:31 kali sudo[120100]: pam_unix(sudo:session): session clo>  
oct 03 13:32:40 kali sudo[125687]:      kali : TTY=pts/1 ; PWD=/home/ka>  
oct 03 13:32:40 kali sudo[125687]: pam_unix(sudo:session): session ope>  
oct 03 13:43:16 kali sudo[131047]:      kali : TTY=pts/3 ; PWD=/home/ka>
```

FASE 5 — Gestión de usuarios y permisos

Objetivo: demostrar creación de cuentas, grupos y buenas prácticas y auditar permisos.

Comandos (ejemplos):

- Crear usuario y grupo:

```
sudo adduser alumno1
```

```
sudo groupadd secops
```

```
sudo usermod -aG secops alumno1
```

```
(kali㉿kali)-[~]
$ sudo adduser alumno1
sudo groupadd secops
sudo usermod -aG secops alumno1

Nueva contraseña:
Vuelva a escribir la nueva contraseña:
passwd: contraseña actualizada correctamente
Cambiando la información de usuario para alumno1
Introduzca el nuevo valor, o pulse INTRO para usar el valor predeterminado
    Nombre completo []: alumno1
    Número de habitación []:
    Teléfono del trabajo []:
    Teléfono de casa []:
    Otro []:
Is the information correct? [Y/n] y

(kali㉿kali)-[~]
$
```

- Asignar permisos a directorio:

```
sudo chown root:secops /srv/seguro
```

```
sudo chmod 750 /srv/seguro
```

- Revisar sudoers (usar visudo):

sudo visudo

```
(kali㉿kali)-[/srv]
$ sudo mkdir seguro

(kali㉿kali)-[/srv]
$ ls
seguro  tftp

(kali㉿kali)-[/srv]
$ sudo chown root:secops /srv/seguro
sudo chmod 750 /srv/seguro

(kali㉿kali)-[/srv]
$
```

```
Defaults    mail_badpass
Defaults    secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"

# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults    use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@include /etc/sudoers.d
```

FASE 6 — Endurecimiento y medidas de protección (firewall, IDS, mitigaciones)

Objetivo: aplicar controles para reducir la superficie de ataque.

Ejemplos de acciones y comandos:

Firewall con nftables (resumen)

- Tabla y cadena básica:

`sudo nft add table inet filtro`

```
(kali㉿kali)-[~]  
$ sudo nft add table inet filtro  
[sudo] contraseña para kali:  
  
(kali㉿kali)-[~]  
$
```

`sudo nft add chain inet filtro input { type filter hook input priority 0 \; policy drop \; }`

```
(kali㉿kali)-[/srv]  
$ sudo nft add chain inet filtro input '{ type filter hook input priority 0; policy drop; }'
```

`sudo nft add rule inet filtro input iif lo accept`

```
(kali㉿kali)-[/srv]  
$ sudo nft add rule inet filtro input iif lo accept
```

`sudo nft add rule inet filtro input tcp dport {22,80,443,3128} accept`

```
(kali㉿kali)-[~]  
$ sudo nft add rule inet filtro input tcp dport '{22,80,443,3128}' accept  
  
(kali㉿kali)-[~]  
$
```

`sudo nft add rule inet filtro input ct state {established,related} accept`

```
(kali㉿kali)-[~]  
$ sudo nft add rule inet filtro input ct state '{established,related}' accept  
  
(kali㉿kali)-[~]  
$
```

- Guardar reglas:

```
sudo nft list ruleset > /etc/nftables.conf
```

```
(root@kali)-[~]  
# sudo nft list ruleset > /etc/nftables.conf  
  
(root@kali)-[~]  
#
```

Firewall con ufw (sencillo)

```
sudo ufw allow 22/tcp
```

```
sudo ufw allow 3128/tcp
```

```
sudo ufw enable
```

```
(root@kali)-[~]  
# sudo ufw allow 22/tcp  
sudo ufw allow 3128/tcp  
sudo ufw enable  
  
Rules updated  
Rules updated (v6)  
Rules updated  
Rules updated (v6)  
Firewall is active and enabled on system startup  
  
(root@kali)-[~]  
#
```


HIPS / Hardening

- Habilitar fail2ban para SSH:

```
sudo apt install fail2ban
```

```
sudo systemctl enable --now fail2ban
```

```
(root@kali)-[~]
# sudo apt install fail2ban
sudo systemctl enable --now fail2ban

Installing:
  fail2ban

Installing dependencies:
  python3-systemd

Paquetes sugeridos:
  mailx  system-log-daemon  monit

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 591
  Download size: 507 kB
  Space needed: 2.461 kB / 8.186 MB available

Continue? [S/n] s
Des:1 http://http.kali.org/kali kali-rolling/main amd64 python3-systemd amd64
235-1+b6 [40,8 kB]
Des:2 http://mirror.es.cdn-perfprod.com/kali kali-rolling/main amd64 fail2ban
all 1.1.0-8 [466 kB]
Descargados 507 kB en 2s (324 kB/s)
Seleccionando el paquete python3-systemd previamente no seleccionado.
(Leyendo la base de datos ... 419125 ficheros o directorios instalados actual
mente.)
Preparando para desempaquetar .../python3-systemd_235-1+b6_amd64.deb ...
Desempaquetando python3-systemd (235-1+b6) ...
Seleccionando el paquete fail2ban previamente no seleccionado.
Preparando para desempaquetar .../fail2ban_1.1.0-8_all.deb ...
Desempaquetando fail2ban (1.1.0-8) ...
Configurando python3-systemd (235-1+b6) ...
Configurando fail2ban (1.1.0-8) ...
update-rc.d: We have no instructions for the fail2ban init script.
update-rc.d: It looks like a network service, we disable it.
fail2ban.service is a disabled or a static unit, not starting it.
Procesando disparadores para kali-menu (2025.3.2) ... ]
```

- Auditar con lynis:

sudo apt install lynis

sudo lynis audit system

```
(kali㉿kali)-[~]
└─$ sudo apt install lynis
sudo lynis audit system

[sudo] contraseña para kali:
Installing:
  lynis

Installing dependencies:
  menu

Paquetes sugeridos:
  apt-listbugs debsecan debsums tripwire samhain aide menu-l10n

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 591
  Download size: 629 kB
  Space needed: 3.282 kB / 8.180 MB available

Continue? [S/n] s
Des:1 http://kali.download/kali kali-rolling/main amd64 lynis all 3.1.4-1 [273 kB]
Des:2 http://kali.download/kali kali-rolling/main amd64 menu amd64 2.1.51 [355 kB]
Descargados 629 kB en 1s (1.214 kB/s)
Seleccionando el paquete lynis previamente no seleccionado.
(Leyendo la base de datos ... 419614 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../archives/lynis_3.1.4-1_all.deb ...
Desempaquetando lynis (3.1.4-1) ...
Seleccionando el paquete menu previamente no seleccionado.
Preparando para desempaquetar .../archives/menu_2.1.51_amd64.deb ...
Desempaquetando menu (2.1.51) ...
Configurando lynis (3.1.4-1) ...
Created symlink '/etc/systemd/system/timers.target.wants/lynis.timer' → '/usr/lib/systemd/system/lynis.timer'.
lynis.service is a disabled or a static unit, not starting it.
Configurando menu (2.1.51) ...
Procesando disparadores para desktop-file-utils (0.28-1) ...
Procesando disparadores para doc-base (0.11.2) ...
Processing 1 added doc-base file...
Procesando disparadores para man-db (2.13.1-1) ...
Procesando disparadores para kali-menu (2025.3.2) ...
Procesando disparadores para menu (2.1.51) ...
```

[Lynis 3.1.4]

```
#####
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.
```

```
2007-2024, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
```

```
#####
```

[+] Initializing program

```
- Detecting OS... [ DONE ]
- Checking profiles... [ DONE ]
- Detecting language and localization [ es ]
```

```
Program version:      3.1.4
Operating system:     linux
Operating system name: Kali Linux
Operating system version: Rolling release
Kernel version:       6.12.38+kali
Hardware platform:    x86_64
Hostname:             kali
```

```
Profiles:             /etc/lynis/default.prf
Log file:              /var/log/lynis.log
Report file:           /var/log/lynis-report.dat
Report version:        1.0
Plugin directory:      /etc/lynis/plugins
```

```
Auditor:              [Not Specified]
Language:              es
Test category:         all
Test group:            all
```

```
- Program update status... [ SIN ACTUALIZACIÓN ]
```

[+] Herramientas del sistema

```
- Scanning available tools...
- Checking system binaries...
```

[+] Plugins (fase 1)

Nota: los plugins contienen pruebas más extensivas y toman más tiempo

```

- Plugin: debian
[
[+] Debian Tests
- Checking for system binaries that are required by Debian Tests...
- Checking /bin... [ FOUND ]
- Checking /sbin... [ FOUND ]
- Checking /usr/bin... [ FOUND ]
- Checking /usr/sbin... [ FOUND ]
- Checking /usr/local/bin... [ FOUND ]
- Checking /usr/local/sbin... [ FOUND ]
- Authentication:
- PAM (Pluggable Authentication Modules):

[WARNING]: Test DEB-0001 had a long execution: 16.931363 seconds

- libpam-tmpdir [ Not Installed ]
- File System Checks:
- DM-Crypt, Cryptsetup & Cryptmount:
- Software:
- apt-listbugs [ Not Installed ]
- apt-listchanges [ Not Installed ]
- needrestart [ Not Installed ]
- fail2ban [ Installed with jail.conf ]
]

[+] Arranque y servicios
- Service Manager [ systemd ]
- Checking UEFI boot [ DESHABILITADO ]
- Checking presence GRUB2 [ ENCONTRADO ]
- Checking for password protection [ NINGUNO ]
- Check running services (systemctl) [ HECHO ]
  Result: found 22 running services
- Check enabled services at boot (systemctl) [ HECHO ]
  Result: found 21 enabled services
- Check startup files (permissions) [ OK ]
- Running 'systemd-analyze security'
  Unit name (exposure value) and predicate

- ModemManager.service (value=6.3) [ MEDIO ]
- NetworkManager.service (value=7.8) [ EXPUESTO ]
- accounts-daemon.service (value=5.5) [ MEDIO ]
- apache2.service (value=9.2) [ INSEGURO ]
- colord.service (value=3.5) [ PROTEGIDO ]
- cron.service (value=9.6) [ INSEGURO ]
- dbus.service (value=9.3) [ INSEGURO ]
- emergency.service (value=9.5) [ INSEGURO ]

```

```

- Checking /var/tmp sticky bit           [ OK ]
- ACL support root file system           [ HABILITADO ]
- Mount options of /                     [ NO POR DEFECTO ]
- Mount options of /dev                  [ PARCIALMENTE BASTIONADO ]
- Mount options of /dev/shm              [ PARCIALMENTE BASTIONADO ]
- Mount options of /run                   [ BASTIONADO ]
- Mount options of /tmp                   [ PARCIALMENTE BASTIONADO ]
- Total without nodev:5 noexec:12 nosuid:3 ro or noexec (W^X): 12 of total 29
- Checking Locate database                [ ENCONTRADO ]
- Disable kernel support of some filesystems

[+] Dispositivos USB
- Checking usb-storage driver (modprobe config) [ NO DESHABILITADO ]
- Checking USB devices authorization           [ HABILITADO ]
- Checking USBGuard                            [ NO ENCONTRADO ]

[+] Almacenamiento
- Checking firewire ohci driver (modprobe config) [ NO DESHABILITADO ]

[+] NFS
- Query rpc registered programs               [ HECHO ]
- Query NFS versions                          [ HECHO ]
- Query NFS protocols                         [ HECHO ]
- Check running NFS daemon                   [ NO ENCONTRADO ]

[+] Servicios de nombres
- Searching DNS domain name                   [ DESCONOCIDO ]
- Checking /etc/hosts
  - Duplicate entries in hosts file           [ NINGUNO ]
  - Presence of configured hostname in /etc/hosts [ ENCONTRADO ]
  - Hostname mapped to localhost              [ NO ENCONTRADO ]
  - Localhost mapping to IP address           [ OK ]

[+] Puertos y paquetes
- Searching package managers
  - Searching RPM package manager             [ ENCONTRADO ]
  - Querying RPM package manager
  - Searching dpkg package manager            [ ENCONTRADO ]
  - Querying package manager

[WARNING]: Test PKGS-7345 had a long execution: 11.624198 seconds

- Query unpurged packages                    [ NINGUNO ]
- Checking APT package database               [ OK ]

```

FASE 7 — Informe final y presentación

Entregables (mínimos):

Se encuentra en el propio documento

Recomendaciones:

- Limitar los puertos expuestos sólo a los necesarios dentro de la red privada
- Implementar una detección temprana de escaneos (con logs detallados del firewall por ejemplo)
- Utilizar sistemas de monitorización continua (Wireshark)
- Mantener actualizado el software del servidor
- Establecer alertas automáticas ante fallos de inicio de sesión o cambios de permisos
- Mantener reglas claras y documentadas del firewall

FASE 1 — Reconocimiento y mapeo de red

Objetivo: identificar hosts activos, puertos y servicios.

Herramientas y comandos (ejemplos):

- Descubrimiento de hosts:
 - arp-scan (si está permitido):
 - sudo arp-scan -l
 - netdiscover:
 - sudo netdiscover -r 192.168.56.0/24
- Escaneo de puertos con nmap:
 - Scan rápido de puertos y servicios:
 - nmap -sS -sV -O -Pn 192.168.56.0/24
 - Escaneo de puertos específicos:
 - nmap -p 22,80,443,3128 192.168.56.101
- Resultado esperado: lista de hosts, puertos abiertos, servicios y versiones.

nmap

```
(kali㉿kali)-[~]  
$ mkdir -p ~/lab/recon ~/lab/recon/nmap_outputs  
  
(kali㉿kali)-[~]  
$ cd ~/lab/recon  
:  
(kali㉿kali)-[~/lab/recon]
```

```
(kali㉿kali)-[~/Escritorio]  
$ nmap -sS -sV -O -Pn 192.168.56.0/24  
  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-03 13:47 CEST  
Nmap scan report for 192.168.56.17  
Host is up (0.0029s latency).  
All 1000 scanned ports on 192.168.56.17 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
MAC Address: 08:00:27:6D:45:BB (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 1 hop  
  
Nmap scan report for 192.168.56.22  
Host is up (0.00074s latency).  
All 1000 scanned ports on 192.168.56.22 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
MAC Address: 08:00:27:86:53:97 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 1 hop  
  
Nmap scan report for 192.168.56.40  
Host is up (0.0020s latency).  
All 1000 scanned ports on 192.168.56.40 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
MAC Address: 08:00:27:75:A0:90 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 1 hop  
  
Nmap scan report for 192.168.56.70  
Host is up (0.0018s latency).  
All 1000 scanned ports on 192.168.56.70 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
MAC Address: 08:00:27:7E:FA:03 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 1 hop  
  
Nmap scan report for 192.168.56.80  
Host is up (0.00075s latency).  
All 1000 scanned ports on 192.168.56.80 are in ignored states.
```

```
(kali㉿kali)-[~/Escritorio]
$ sudo arp-scan -l

[sudo] contraseña para kali:
Interface: eth0, type: EN10MB, MAC: 08:00:27:c7:b6:ff, IPv4: 10.0.2.15
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
10.0.2.2      52:55:0a:00:02:02      (Unknown: locally administered)
10.0.2.3      52:55:0a:00:02:03      (Unknown: locally administered)

2 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.958 seconds (130.75 hosts/sec). 2 responded
```

```
Session  Acciones  Editar  Vista  Ayuda
Currently scanning: Finished! | Screen View: Unique Hosts

0 Captured ARP Req/Rep packets, from 0 hosts. Total size: 0

+-----+-----+-----+-----+-----+-----+
| IP           | At MAC Address | Count | Len | MAC Vendor / Hostname |
+-----+-----+-----+-----+-----+-----+
|               |                 |       |     |                       |
+-----+-----+-----+-----+-----+-----+
```

```
(kali㉿kali)-[~/Escritorio]
$ nmap -p 22,80,443,3128 192.168.56.101

Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-03 13:54 CEST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.46 seconds
```

Tareas del equipo:

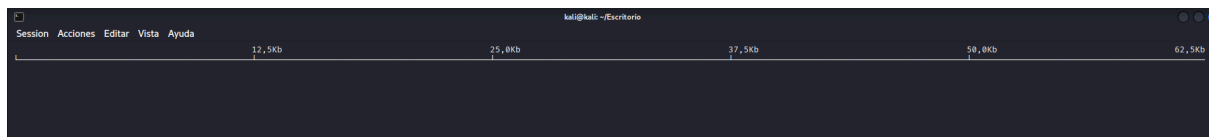
- Recon/Red hace mapeo y sube hosts.csv con IP, hostname, puertos y servicios detectados.
- Indicar servicios críticos (SSH, HTTP, RDP, DB, proxy).

FASE 2 — Análisis de tráfico y paquetes

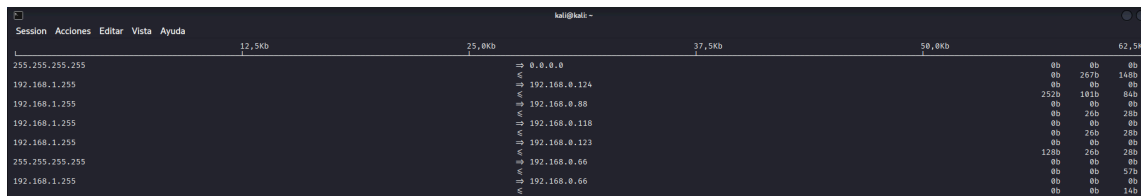
Objetivo: capturar tráfico relevante y analizar patrones/anomalías.

Herramientas y comandos:

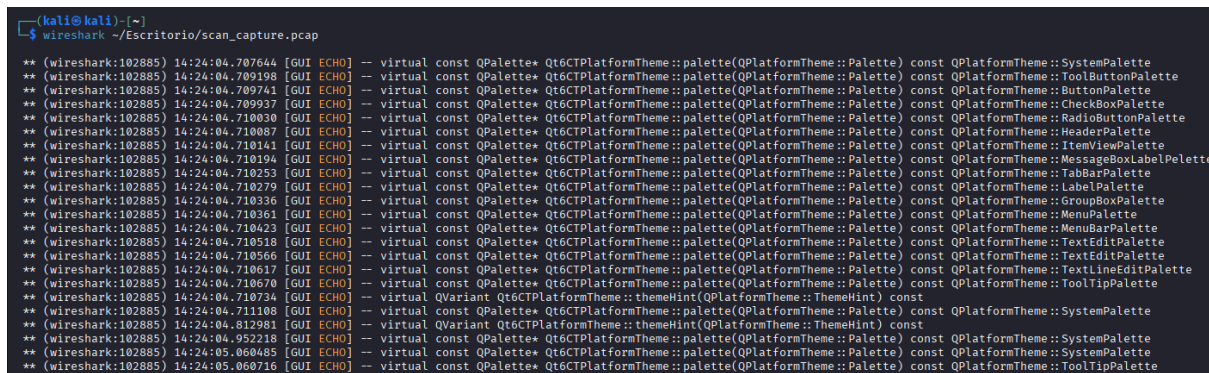
- Captura de paquetes:
 - tcpdump para capturar en interfaz:
 - `sudo tcpdump -i eth0 -w /home/kali/pcaps/scan_capture.pcap`
 - Wireshark
- Monitoreo en vivo:
 - iftop (consumo por conx):
 - `sudo iftop -i eth0`
 - nethogs (por proceso)



-
- Análisis de .pcap:
 - Abrir pcap con **Wireshark** (GUI) o tshark:



-
- tshark -r scan_capture.pcap -q -z io,phs



-
- Extracción de flujos/estadísticas con pyshark / scripts Python (opcional).

Tareas del equipo:

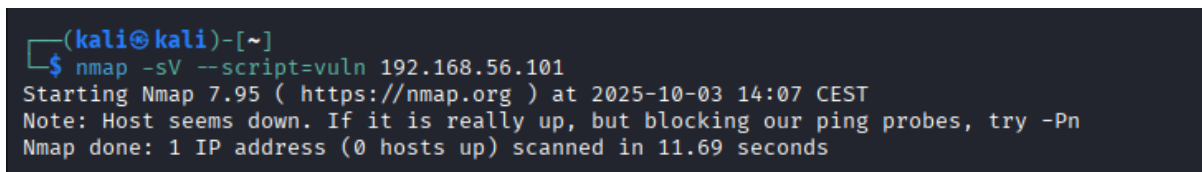
- Capturar tráfico durante un periodo de actividad (p. ej. navegación desde cliente a través de proxy).
- Extraer 3 flujos relevantes y explicar (IP origen/destino, puertos, bytes, flags).
- Detectar posibles anomalías (tráfico inesperado, puertos inusuales).

FASE 3 — Enumeración de servicios y pruebas seguras

Objetivo: verificar configuraciones visibles (vulnerabilidades de servicio *sin explotar*).

Herramientas:

- nmap scripts (--script), ss, curl, nikto (solo pruebas no destructivas).
- Ejemplos:
- nmap -sV --script=vuln 192.168.56.101



- curl -I <http://192.168.56.101:80>


```
(kali㉿kali)-[~]
$ curl -I http://192.168.56.101:80
curl: (7) Failed to connect to 192.168.56.101 port 80 after 3058 ms: Could not connect to server
```

- nikto -host <http://192.168.56.101:80>

```
(kali㉿kali)-[~]
$ nikto -host http://192.168.56.101:80

- Nikto v2.5.0
_____
+ 0 host(s) tested
```

-

Nota: no realizar explotación sin autorización.

Tareas:

- Comprobar encabezados HTTP, existencia de directorios sensibles, versiones de servicios.
- Anotar hallazgos: servicios sin TLS, versiones obsoletas, páginas de administración accesibles, etc.

FASE 4 — Análisis de logs y forense básico

Objetivo: revisar logs del sistema/servicios para identificar actividad sospechosa.

Ubicaciones y herramientas:

- Logs de sistema: /var/log/syslog, /var/log/auth.log
- .

```
(kali㉿kali)-[/var/log]
$ sudo tail -n 200 /var/log/auth.log

a

(kali㉿kali)-[/var/log]
$
```

```
(kali㉿kali)-[/var/log]
$ sudo grep "Failed password" /var/log/auth.log | wc -l
0
```

-
- Logs de aplicaciones: Squid /var/log/squid/access.log, web /var/log/apache2/access.log.
- Herramientas: grep, ausearch(auditd), journalctl, ELK/Graylog (si disponible).
- Comandos útiles:
- sudo tail -n 200 /var/log/auth.log
- sudo grep "Failed password" /var/log/auth.log | wc -l
- sudo journalctl -u ssh

```
(kali@kali)-[/var/log]
$ sudo journalctl -u ssh
[sudo] contraseña para kali:
oct 03 09:06:22 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
oct 03 09:06:22 kali sshd[5968]: Server listening on 0.0.0.0 port 22.
oct 03 09:06:22 kali sshd[5968]: Server listening on :: port 22.
oct 03 09:06:22 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
oct 03 09:07:12 kali sshd-session[6398]: Accepted password for kali from 192.168.0.26 port 49991 ssh2
oct 03 09:11:32 kali systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
oct 03 09:11:32 kali systemd[1]: ssh.service: Deactivated successfully.
oct 03 09:11:32 kali systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
-- Boot f767bdacac72415882b99b1d1ea05b16 --
oct 03 09:12:20 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
oct 03 09:12:20 kali sshd[1809]: Server listening on 0.0.0.0 port 22.
oct 03 09:12:20 kali sshd[1809]: Server listening on :: port 22.
oct 03 09:12:20 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
oct 03 09:12:24 kali unix_chkpwd[1854]: password check failed for user (kali)
oct 03 09:12:24 kali sshd-session[1836]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.0.26 user=kali
oct 03 09:12:24 kali sshd-session[1836]: pam_winbind(sshd:auth): getting password (0x00000388)
oct 03 09:12:24 kali sshd-session[1836]: pam_winbind(sshd:auth): pam_get_item returned a password
oct 03 09:12:24 kali sshd-session[1836]: pam_winbind(sshd:auth): request wbcLogonUser failed: WBC_ERR_WINBIND_NOT_AVAILABLE, PAM error: PAM_AUTHINFO_UNAVAIL (9):
oct 03 09:12:24 kali sshd-session[1836]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'kali')
oct 03 09:12:26 kali sshd-session[1836]: Failed password for kali from 192.168.0.26 port 50022 ssh2
oct 03 09:12:30 kali sshd-session[1836]: Accepted password for kali from 192.168.0.26 port 50022 ssh2
oct 03 09:12:30 kali sshd-session[1836]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
```

Tareas:

- Identificar intentos fallidos de login, accesos a URLs no autorizadas, timestamps correlacionados con capturas pcap.
- Generar un timeline básico (CSV) con evento, hora, IP origen, descripción.

FASE 5 — Gestión de usuarios y permisos

Objetivo: demostrar creación de cuentas, grupos y buenas prácticas y auditar permisos.

Comandos (ejemplos):

- Crear usuario y grupo:

```
(kali@kali)-[/]
$ sudo adduser usuario1
Nueva contraseña:
Vuelva a escribir la nueva contraseña:
passwd: contraseña actualizada correct
Cambiando la información de usuario pa
Introduzca el nuevo valor, o pulse INT
Nombre completo []:
Número de habitación []:
Teléfono del trabajo []:
Teléfono de casa []:
Otro []:
Is the information correct? [Y/n] y
```

- sudo adduser alumno1
- sudo groupadd secops

```
(kali@kali)-[/]
$ sudo groupadd secops
```

- sudo usermod -aG secops alumno1

```
(kali@kali)-[/]
$ sudo usermod -aG secops usuario1
```

- Asignar permisos a directorio:
- `sudo chown root:secops /srv/seguro`

```
(kali㉿kali)-[/srv]
$ sudo chown root:secops /srv/seguro
```

-
- `sudo chmod 750 /srv/seguro`

```
(kali㉿kali)-[/srv]
$ sudo chmod 750 /srv/seguro
```

-
- Revisar sudoers (usar visudo):
- `sudo visudo`

```
# User privilege specification
root    ALL=(ALL:ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d
```

Tareas:

- Crear 1 usuario por sub-equipo, asignar grupo y dar acceso restringido a un directorio.
- Documentar comandos y verificar con `id` usuario y `ls -ld`.

FASE 6 — Endurecimiento y medidas de protección (firewall, IDS, mitigaciones)

Objetivo: aplicar controles para reducir la superficie de ataque.

Ejemplos de acciones y comandos:

Firewall con nftables (resumen)

- Tabla y cadena básica:

```
(kali㉿kali)-[/srv]
$ sudo nft add table inet filtro

(kali㉿kali)-[/srv]
$
```

- `sudo nft add table inet filtro`
- `sudo nft add chain inet filtro input { type filter hook input priority 0 \; policy drop \; }`

- ```
(kali㉿kali)-[/srv]
$ sudo nft add chain inet filtro input '{ type filter hook input priority 0; policy drop ; }'
```
- - sudo nft add rule inet filtro input iif lo accept
- ```
(kali㉿kali)-[/srv]
$ sudo nft add rule inet filtro input iif lo accept

(kali㉿kali)-[/srv]
$
```
- - sudo nft add rule inet filtro input tcp dport {22,80,443,3128} accept
- ```
(kali㉿kali)-[/]
$ sudo nft add rule inet filtro input tcp dport '{22,80,443,3128}' accept

(kali㉿kali)-[/]
$
```
- - sudo nft add rule inet filtro input ct state {established,related} accept
- ```
(kali㉿kali)-[/]
$ sudo nft add rule inet filtro input ct state '{established,related}' accept

(kali㉿kali)-[/]
$
```
- - Guardar reglas:
 - sudo nft list ruleset > /etc/nftables.conf
- ```
(kali㉿kali)-[/]
$ sudo nft list ruleset > /etc/nftables.conf
zsh: permiso denegado: /etc/nftables.conf

(kali㉿kali)-[/]
$ sudo su
(kali㉿kali)-[/]
sudo nft list ruleset > /etc/nftables.conf

(kali㉿kali)-[/]
#
```
- 

### Firewall con ufw (sencillo)

sudo ufw allow 22/tcp

```
(kali㉿kali)-[/]
$ sudo ufw allow 22/tcp
Rules updated
Rules updated (v6)

(kali㉿kali)-[/]
```

sudo ufw allow 3128/tcp

```
(kali㉿kali)-[/]
$ sudo ufw allow 3128/tcp

Rules updated
Rules updated (v6)
```

sudo ufw enable

```
(kali㉿kali)-[/]
$ sudo ufw enable
Firewall is active and enabled on system startup

(kali㉿kali)-[/]
$
```

## HIPS / Hardening

- Habilitar fail2ban para SSH:
- sudo apt install fail2ban

```
(kali㉿kali)-[/]
$ sudo apt install fail2ban
Installing:
 fail2ban

Installing dependencies:
 python3-systemd

Paquetes sugeridos:
 mailx system-log-daemon monit

Summary:
 Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 3
 Download size: 507 kB
 Space needed: 2.461 kB / 3.872 MB available
```

```
Ign:1 http://http.kali.org/kali kali-rolling/main amd64 python3-systemd amd64 235-1+b6
Ign:2 http://http.kali.org/kali kali-rolling/main amd64 fail2ban all 1.1.0-8
Ign:1 http://http.kali.org/kali kali-rolling/main amd64 python3-systemd amd64 235-1+b6
Ign:2 http://http.kali.org/kali kali-rolling/main amd64 fail2ban all 1.1.0-8
```

- 0% [Trabajando]
- sudo systemctl enable --now fail2ban

```
(kali@kali)-[/]
$ sudo systemctl enable --now fail2ban
Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
Created symlink '/etc/systemd/system/multi-user.target.wants/fail2ban.service' → '/usr/lib/systemd/system/fail2ban.service'.
```

- 
- Auditar con lynis:
- sudo apt install lynis

```
(kali@kali)-[~]
$ sudo apt install lynis
Installing:
 lynis

Installing dependencies:
 menu

Paquetes sugeridos:
 apt-listbugs debsecan debsums tripwire samhain aide fail2ban menu-l10n

Summary:
 Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 3
 Download size: 629 kB
 Space needed: 3.282 kB / 3.875 MB available

Continue? [S/n] s
Des:1 http://kali.download/kali kali-rolling/main amd64 lynis all 3.1.4-1 [273 kB]
Des:2 http://kali.download/kali kali-rolling/main amd64 menu amd64 2.1.51 [355 kB]
Descargados 629 kB en 1s (1.024 kB/s)
Seleccionando el paquete lynis previamente no seleccionado.
(Leyendo la base de datos ... 434336 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../archives/lynis_3.1.4-1_all.deb ...
Desempaquetando lynis (3.1.4-1) ...
Seleccionando el paquete menu previamente no seleccionado.
Preparando para desempaquetar .../archives/menu_2.1.51_amd64.deb ...
Desempaquetando menu (2.1.51) ...
Configurando lynis (3.1.4-1) ...
Created symlink '/etc/systemd/system/timers.target.wants/lynis.timer' → '/usr/lib/systemd/system/lynis.timer'.
lynis.service is a disabled or a static unit, not starting it.
Configurando menu (2.1.51) ...
Procesando disparadores para desktop-file-utils (0.28-1) ...
Procesando disparadores para doc-base (0.11.2) ...
Processing 1 added doc-base file...
Procesando disparadores para man-db (2.13.1-1) ...
Procesando disparadores para kali-menu (2025.4.1) ...
Procesando disparadores para menu (2.1.51) ...
```

- 
- sudo lynis audit system

```
(kali㉿kali)-[~]
$ sudo lynis audit system

[Lynis 3.1.4]

#####
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.

2007-2024, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
#####

[+] Initializing program

- Detecting OS ... [DONE]
- Checking profiles ... [DONE]
- Detecting language and localization [es]

Program version: 3.1.4
Operating system: Linux
Operating system name: Kali Linux
Operating system version: Rolling release
Kernel version: 6.12.38+kali
Hardware platform: x86_64
Hostname: kali

Profiles: /etc/lynis/default.prf
Log file: /var/log/lynis.log
Report file: /var/log/lynis-report.dat
Report version: 1.0
Plugin directory: /etc/lynis/plugins

Auditor: [Not Specified]
Language: es
```



**[+] Debian Tests**

```

- Checking for system binaries that are required by Debian Tests ...
- Checking /bin ... [FOUND]
- Checking /sbin ... [FOUND]
- Checking /usr/bin ... [FOUND]
- Checking /usr/sbin ... [FOUND]
- Checking /usr/local/bin ... [FOUND]
- Checking /usr/local/sbin ... [FOUND]
- Authentication:
- PAM (Pluggable Authentication Modules):
 - libpam-tmpdir [Not Installed]
- File System Checks:
- DM-Crypt, Cryptsetup & Cryptmount:
- Software:
- apt-listbugs [Not Installed]
- apt-listchanges [Not Installed]
- needrestart [Not Installed]
- fail2ban [Not Installed]
]

```

**[+] Arranque y servicios**

```

- Service Manager [systemd]
- Checking UEFI boot [DESHABILITADO]
- Checking presence GRUB2 [ENCONTRADO]
- Checking for password protection [NINGUNO]
- Check running services (systemctl) [HECHO]
 Result: found 18 running services
- Check enabled services at boot (systemctl) [HECHO]
 Result: found 19 enabled services
- Check startup files (permissions) [OK]
- Running 'systemd-analyze security'
 Unit name (exposure value) and predicate

- ModemManager.service (value=6.3) [MEDIO]
- NetworkManager.service (value=7.8) [EXPUESTO]
- accounts-daemon.service (value=5.5) [MEDIO]
- colord.service (value=3.5) [PROTEGIDO]
- cron.service (value=9.6) [INSEGURO]
- dbus.service (value=9.3) [INSEGURO]
- emergency.service (value=9.5) [INSEGURO]
- getty@tty1.service (value=9.6) [INSEGURO]
- haveged.service (value=3.2) [PROTEGIDO]
- lightdm.service (value=9.6) [INSEGURO]
- lynis.service (value=9.6) [INSEGURO]
- pcscd.service (value=2.1) [PROTEGIDO]
- plymouth-start.service (value=9.5) [INSEGURO]

```

```
[+] Servicios inseguros
- Installed inetd package [NO ENCONTRADO]
- Installed xinetd package [OK]
 - xinetd status [SIN ACTIVAR]
- Installed rsh client package [SUGERENCIA]
- Installed rsh server package [OK]
- Installed telnet client package [OK]
- Installed telnet server package [NO ENCONTRADO]
- Checking NIS client installation [OK]
- Checking NIS server installation [OK]
- Checking TFTP client installation [SUGERENCIA]
- Checking TFTP server installation [SUGERENCIA]

[+] Banners e identificación
- /etc/issue [ENCONTRADO]
- /etc/issue contents [DÉBIL]
- /etc/issue.net [ENCONTRADO]
- /etc/issue.net contents [DÉBIL]

[+] Tareas programadas
- Checking crontab and cronjob files [HECHO]

[+] Contabilidad
- Checking accounting information [NO ENCONTRADO]
- Checking sysstat accounting data [DESHABILITADO]
- Checking auditd [NO ENCONTRADO]

[+] Tiempo y sincronización
```

#### Lynis security scan details:

```
Hardening index : 62 [#####]
Tests performed : 272
Plugins enabled : 1
```

#### Components:

```
- Firewall [V]
- Malware scanner [X]
```

#### Scan mode:

```
Normal [V] Forensics [] Integration [] Pentest []
```

#### Lynis modules:

```
- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]
```

#### Files:

```
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat
```

#### Lynis 3.1.4

```
Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)
```

```
2007-2024, CISOfy - https://cisofy.com/lynis/
```

```
Enterprise support available (compliance, plugins, interface and tools)
```

```
[TIP]: Enhance Lynis audits by adding your settings to custom.prfl (see /etc/lynis/default.prfl for all settings)
```

#### Tareas:

- Aplicar reglas mínimas de firewall que permitan solo lo necesario.
- Activar fail2ban y demostrar que bloquea after X intentos (simulación).

- Documentar configuración y justificar decisiones.

## FASE 7 — Informe final y presentación

### Entregables (mínimos):

1. report/Informe\_EquipoX.pdf con:
  - Alcance y reglas del laboratorio.
  - Resumen de hallazgos (hosts, puertos, servicios).
  - Evidencias (pcap snippets, logs extractados, capturas).
  - Riesgos priorizados y medidas de remediación (con comandos).
  - Cronograma y roles.
2. Carpeta pcaps/, logs/, scripts/.
3. Diapositiva para presentación

### Herramientas del módulo «Seguridad informática» (recordatorio para la práctica)

- Recon / Scanning: nmap, netdiscover, arp-scan, masscan (con precaución).
- Captura / Análisis de tráfico: tcpdump, tshark, Wireshark, iftop, nethogs.
- Packet crafting / análisis: scapy, pyshark.
- Logs / SIEM básico: journalctl, auditd, rsyslog, ELK (Elasticsearch + Logstash + Kibana) o Splunk (si disponible).
- Forense / integridad: sleuthkit, autopsy, chkrootkit, rkhunter.
- Hardening / auditoría: ufw, nftables, iptables (legacy), fail2ban, lynis.
- Proxy / Web: squid, apache2, nginx, nikto.
- Gestión de usuarios/permisos: adduser, groupadd, usermod, chmod, chown, visudo.
- Scripting y automatización: Bash, Python (pandas, scikit-learn opcional), Streamlit (visual).
- Documentación / versiones: git.

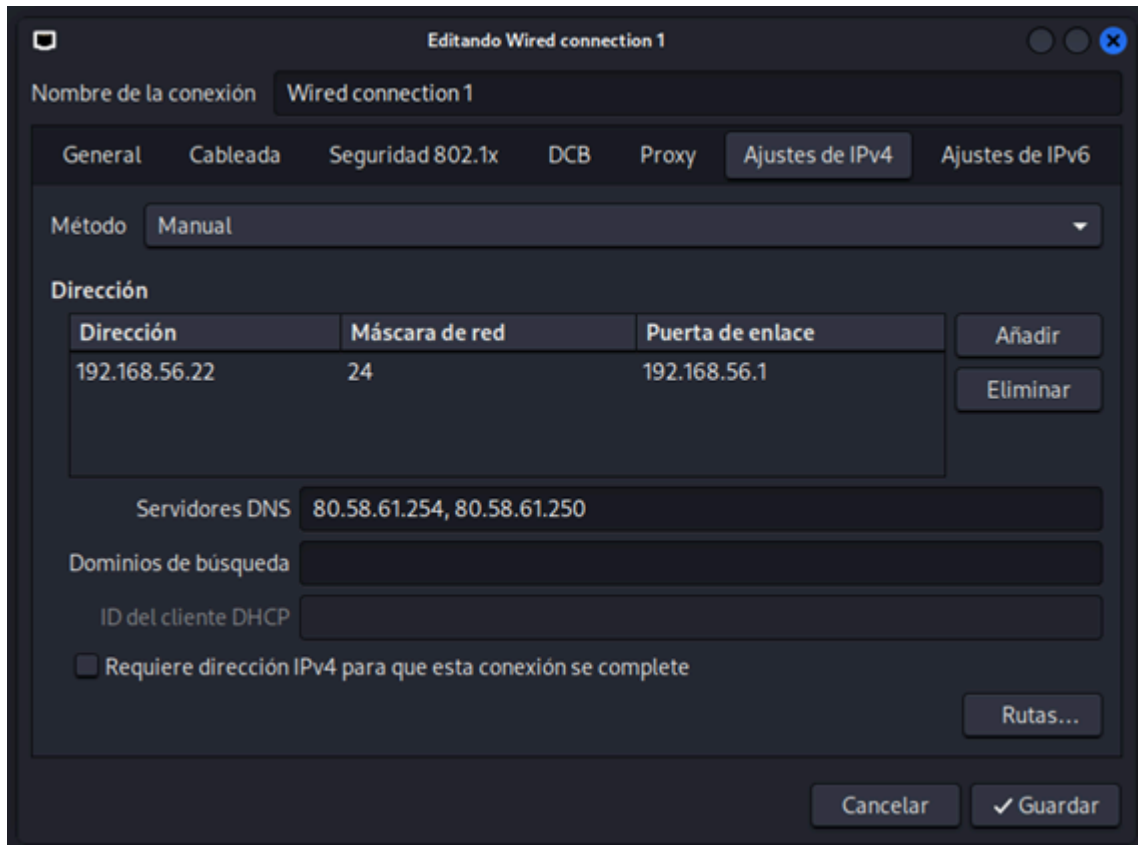
### Ejercicios concretos para entregar (lista rápida)

1. recon/hosts.csv + nmap output.
2. pcaps/pcap1.pcap con explicación de 3 flujos.
3. logs/logs\_issue.txt con 3 hallazgos importantes extraídos de logs.
4. hardening/ con nftables.conf y pasos ejecutados (comandos).
5. report/Informe\_EquipoX.pdf y 5–8 diapositivas resumen.

Optativo quien quiera realizar una exposición en grupo del trabajo realizado o al menos explicar al resto de compañeros lo que han realizado.

**FASE 0 —.**

- Definir ordenadores/segmentos IP (ej. 192.168.56.0/24).
- Documentar alcance y reglas (qué está permitido).
- Crear repositorio con estructura: recon/, pcaps/, logs/, report/.



## FASE 1 — Reconocimiento y mapeo de red

**Objetivo:** identificar hosts activos, puertos y servicios.

### Herramientas y comandos (ejemplos):

- Descubrimiento de hosts:
  - arp-scan (si está permitido):
    - sudo arp-scan -l
  - netdiscover:
    - sudo netdiscover -r 192.168.56.0/24

```
(kali㉿kali)-[~]
└─$ sudo arp-scan -l
Interface: eth0, type: EN10MB, MAC: 08:00:27:86:53:97, IPv4: 192.168.56.22
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.56.17 08:00:27:6d:45:bb (Unknown)
192.168.56.40 08:00:27:75:a0:90 (Unknown)
192.168.56.70 08:00:27:7e:fa:03 (Unknown)
192.168.56.75 08:00:27:5f:6d:09 (Unknown)
192.168.56.80 08:00:27:61:0f:54 (Unknown)

5 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.806 seconds (141.75 hosts/sec). 5 responded

(kali㉿kali)-[~]
└─$ sudo netdiscover -r 192.168.56.0/24
```

kali@kali: ~

Session Acciones Editar Vista Ayuda

Currently scanning: Finished! | Screen View: Unique Hosts

1034 Captured ARP Req/Rep packets, from 23 hosts. Total size: 62040

| IP            | At MAC Address    | Count | Len   | MAC Vendor / Hostname                       |
|---------------|-------------------|-------|-------|---------------------------------------------|
| 192.168.0.115 | 18:60:24:e9:3b:2a | 61    | 3660  | Hewlett Packard                             |
| 192.168.0.65  | e8:39:35:57:9e:e9 | 13    | 780   | Hewlett Packard                             |
| 192.168.56.17 | 08:00:27:6d:45:bb | 1     | 60    | PCS Systemtechnik GmbH                      |
| 192.168.56.40 | 08:00:27:75:a0:90 | 1     | 60    | PCS Systemtechnik GmbH                      |
| 192.168.56.70 | 08:00:27:7e:fa:03 | 24    | 1440  | PCS Systemtechnik GmbH                      |
| 192.168.56.75 | 08:00:27:5f:6d:09 | 1     | 60    | PCS Systemtechnik GmbH                      |
| 192.168.56.80 | 08:00:27:61:0f:54 | 4     | 240   | PCS Systemtechnik GmbH                      |
| 192.168.0.74  | 18:60:24:f5:a4:18 | 6     | 360   | Hewlett Packard                             |
| 192.168.0.123 | 34:17:eb:c4:6c:81 | 97    | 5820  | Dell Inc.                                   |
| 192.168.0.55  | 08:ed:ed:b5:aa:e6 | 2     | 120   | Zhejiang Dahua Technology Co., Ltd.         |
| 192.168.0.52  | d8:43:ae:44:e2:69 | 13    | 780   | Micro-Star INTL CO., LTD.                   |
| 192.168.0.90  | a4:97:33:4e:9f:c4 | 705   | 42300 | ASKEY COMPUTER CORP                         |
| 192.168.0.24  | d8:43:ae:44:e2:72 | 13    | 780   | Micro-Star INTL CO., LTD.                   |
| 192.168.0.78  | dc:4a:3e:8d:c1:b5 | 10    | 600   | Hewlett Packard                             |
| 192.168.0.85  | 08:ed:ed:b6:33:cc | 2     | 120   | Zhejiang Dahua Technology Co., Ltd.         |
| 192.168.0.42  | d8:43:ae:44:e1:d8 | 10    | 600   | Micro-Star INTL CO., LTD.                   |
| 192.168.0.73  | dc:4a:3e:7c:25:3e | 30    | 1800  | Hewlett Packard                             |
| 0.0.0.0       | 08:00:27:61:0f:54 | 6     | 360   | PCS Systemtechnik GmbH                      |
| 192.168.0.100 | 50:c2:e8:d1:19:6f | 1     | 60    | CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD |
| 192.168.0.91  | 08:00:27:61:0f:54 | 3     | 180   | PCS Systemtechnik GmbH                      |
| 192.168.0.36  | 6c:62:6d:87:2a:2c | 10    | 600   | Micro-Star INT'L CO., LTD                   |
| 192.168.0.67  | 18:60:24:e9:3a:50 | 20    | 1200  | Hewlett Packard                             |
| 192.168.0.48  | 08:00:27:6f:98:bb | 1     | 60    | PCS Systemtechnik GmbH                      |

- Escaneo de puertos con nmap:
  - Scan rápido de puertos y servicios:
  - `nmap -sS -sV -O -Pn 192.168.56.0/24`

```
(kali@kali)-[~]
$ nmap -sS -sV -O -Pn 192.168.56.0/24
Starting Nmap 7.95 (https://nmap.org) at 2025-10-03 14:11 CEST
Nmap scan report for 192.168.56.17
Host is up (0.0017s latency).
All 1000 scanned ports on 192.168.56.17 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:6D:45:BB (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

Nmap scan report for 192.168.56.40
Host is up (0.00080s latency).
All 1000 scanned ports on 192.168.56.40 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:75:A0:90 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

Nmap scan report for 192.168.56.70
Host is up (0.00076s latency).
All 1000 scanned ports on 192.168.56.70 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:7E:FA:03 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

Nmap scan report for 192.168.56.75
Host is up (0.00083s latency).
All 1000 scanned ports on 192.168.56.75 are in ignored states.
```

```
Nmap scan report for 192.168.56.80
Host is up (0.00085s latency).
All 1000 scanned ports on 192.168.56.80 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:61:0F:54 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

Nmap scan report for 192.168.56.22
Host is up (0.000050s latency).
All 1000 scanned ports on 192.168.56.22 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (6 hosts up) scanned in 31.23 seconds
```



- Escaneo de puertos específicos:
- `nmap -p 22,80,443,3128 192.168.56.75`
- Resultado esperado: lista de hosts, puertos abiertos, servicios y versiones.

```
(kali㉿kali)-[~]
$ nmap -p 22,80,443,3128 192.168.56.75
Starting Nmap 7.95 (https://nmap.org) at 2025-10-03 14:16 CEST
Nmap scan report for 192.168.56.75
Host is up (0.00076s latency).

PORT STATE SERVICE
22/tcp closed ssh
80/tcp closed http
443/tcp closed https
3128/tcp closed squid-http
MAC Address: 08:00:27:5F:6D:09 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

### Tareas del equipo:

- Recon/Red hace mapeo y sube hosts.csv con IP, hostname, puertos y servicios detectados.
  - Indicar servicios críticos (SSH, HTTP, RDP, DB, proxy).
- 

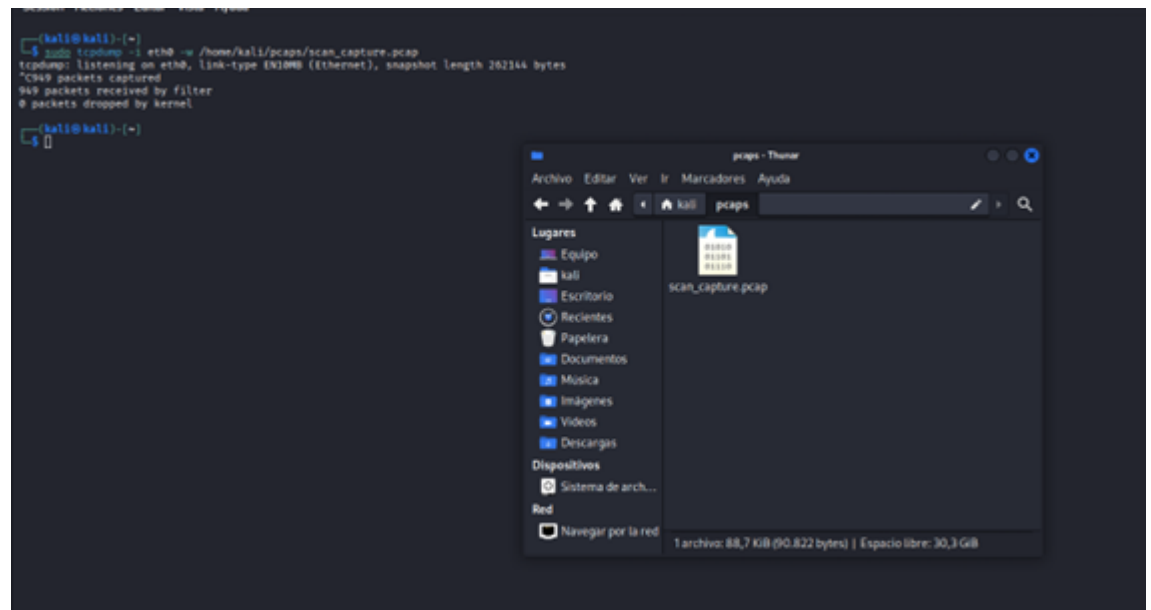
## FASE 2 — Análisis de tráfico y paquetes

**Objetivo:** capturar tráfico relevante y analizar patrones/anomalías.

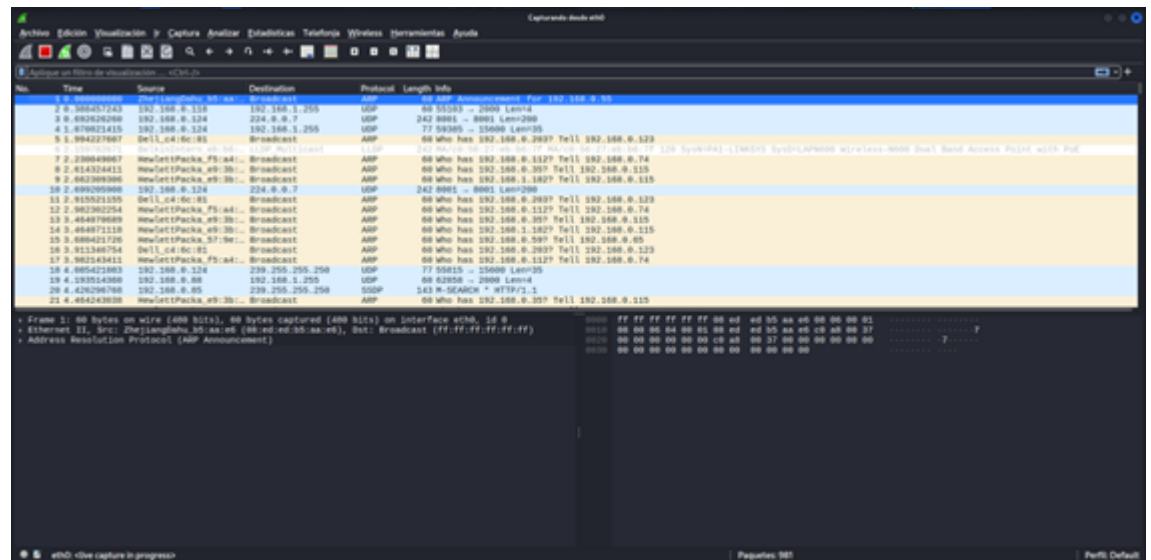
### Herramientas y comandos:

- Captura de paquetes:
  - `tcpdump` para capturar en interfaz:  
  
`sudo tcpdump -i eth0 -w /home/kali/pcaps/scan_capture.pcap`





- Wireshark



- Monitoreo en vivo:
  - iftop (consumo por conx):

sudo iftop -i eth0

```
(kali㉿kali)-[~]
$ sudo iftop -i eth0
[sudo] contraseña para kali:
interface: eth0
IP address is: 192.168.56.22
MAC address is: 08:00:27:86:53:97

(kali㉿kali)-[~]
$
```

The screenshot shows the iftop utility running on interface eth0. It displays a table of active connections with columns for local IP, remote IP, local port, remote port, and data transfer rates. The connections are between 192.168.1.215 and 192.168.1.216 on ports 215 and 216. The data transfer rates are shown in MB/s.

| Local IP      | Remote IP     | Local Port | Remote Port | Local Rate | Remote Rate |
|---------------|---------------|------------|-------------|------------|-------------|
| 192.168.1.215 | 192.168.1.216 | 215        | 216         | 1000       | 1000        |
| 192.168.1.215 | 192.168.1.216 | 215        | 216         | 1000       | 1000        |
| 192.168.1.215 | 192.168.1.216 | 215        | 216         | 1000       | 1000        |
| 192.168.1.215 | 192.168.1.216 | 215        | 216         | 1000       | 1000        |
| 192.168.1.215 | 192.168.1.216 | 215        | 216         | 1000       | 1000        |
| 192.168.1.215 | 192.168.1.216 | 215        | 216         | 1000       | 1000        |

sudo nethogs (por proceso)

The screenshot shows the nethogs utility running on interface eth0. It displays a table of network traffic by process. The processes are listed in the first column, and the network traffic is shown in the second column. The data transfer rates are shown in MB/s.

| Process | Local Rate | Remote Rate |
|---------|------------|-------------|
| sshd    | 1000       | 1000        |
| sshd    | 1000       | 1000        |
| sshd    | 1000       | 1000        |
| sshd    | 1000       | 1000        |
| sshd    | 1000       | 1000        |
| sshd    | 1000       | 1000        |

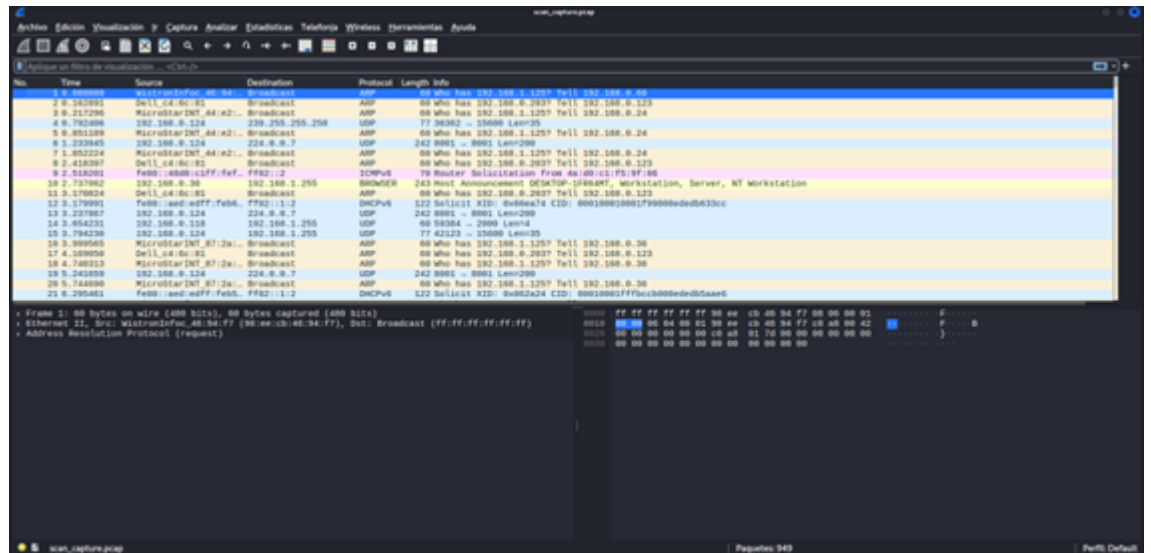


- Análisis de .pcap:
  - Abrir pcap con **Wireshark** (GUI) o tshark:

```

kali@kali: ~/pcaps
$ wireshark -K -i eth0 -s 0.0.0.0/0.0.0.0 -f 'eth0 and ip.addr == 10.0.0.1'
wireshark: 15014 09:38:00.435449 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::SystemPalette
wireshark: 15014 09:38:00.436162 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::ToolButtonPalette
wireshark: 15014 09:38:00.436629 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::ButtonPalette
wireshark: 15014 09:38:00.436843 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::CheckBoxPalette
wireshark: 15014 09:38:00.436945 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::RadioButtonPalette
wireshark: 15014 09:38:00.437011 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::HeaderPalette
wireshark: 15014 09:38:00.437039 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::ItemViewPalette
wireshark: 15014 09:38:00.437387 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::MessageBoxLabelPalette
wireshark: 15014 09:38:00.437396 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::TabBarPalette
wireshark: 15014 09:38:00.437220 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::LabelPalette
wireshark: 15014 09:38:00.437267 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::GroupBoxPalette
wireshark: 15014 09:38:00.437360 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::MenuBarPalette
wireshark: 15014 09:38:00.437388 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::MenuBarPalette
wireshark: 15014 09:38:00.437451 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::TabBarPalette
wireshark: 15014 09:38:00.437478 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::TextEditPalette
wireshark: 15014 09:38:00.437544 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::TextLineEditPalette
wireshark: 15014 09:38:00.437571 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::ToolTipPalette
wireshark: 15014 09:38:00.437644 [GUI] ECHO -- virtual QVariant Q6CTPlatformTheme::themeHint(QPlatformTheme::ThemeHint) const
wireshark: 15014 09:38:00.438000 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::SystemPalette
wireshark: 15014 09:38:00.438070 [GUI] ECHO -- virtual QVariant Q6CTPlatformTheme::themeHint(QPlatformTheme::ThemeHint) const
wireshark: 15014 09:38:00.438077 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::SystemPalette
wireshark: 15014 09:38:00.438087 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::SystemPalette
wireshark: 15014 09:38:00.438087 [GUI] ECHO -- virtual const QPalette* Q6CTPlatformTheme::palette(QPlatformTheme::Palette) const QPlatformTheme::ToolTipPalette

```



- tshark -r scan\_capture.pcap -q -z io,phs

```
(kali@kali)-[~]
└─$ tshark -r scan_capture.pcap -q -z io,phs
tshark: The file "scan_capture.pcap" doesn't exist.

(kali@kali)-[~]
└─$ ls
Descargas Documentos Escritorio Imágenes Música pcaps Plantillas Público Videos

(kali@kali)-[~]
└─$ cd pcaps

(kali@kali)-[~/pcaps]
└─$ ls
scan_capture.pcap wireshark.pcapng

(kali@kali)-[~/pcaps]
└─$ tshark -r scan_capture.pcap -q -z io,phs

Protocol Hierarchy Statistics
Filter:

eth frames:949 bytes:75614
arp frames:653 bytes:39180
ip frames:225 bytes:27477
udp frames:206 bytes:26337
 data frames:177 bytes:21640
 nbdm frames:1 bytes:243
 smb frames:1 bytes:243
 mailslot frames:1 bytes:243
 browser frames:1 bytes:243
 sdp frames:7 bytes:1001
 mdns frames:21 bytes:3453
 igmp frames:19 bytes:1140
 ipv6 frames:67 bytes:7989
 icmpv6 frames:38 bytes:3140
 udp frames:29 bytes:4849
 dhcpv6 frames:8 bytes:976
 mdns frames:21 bytes:3873
 lldp frames:4 bytes:968

(kali@kali)-[~/pcaps]
└─$
```

- Extracción de flujos/estadísticas con pyshark / scripts Python (opcional).

## Tareas del equipo:

- Capturar tráfico durante un periodo de actividad (p. ej. navegación desde cliente a través de proxy).
  - Extraer 3 flujos relevantes y explicar (IP origen/destino, puertos, bytes, flags).
  - Detectar posibles anomalías (tráfico inesperado, puertos inusuales).
- 

### FASE 3 — Enumeración de servicios y pruebas seguras

**Objetivo:** verificar configuraciones visibles (vulnerabilidades de servicio *sin explotar*).

**Herramientas:**

- nmap scripts (--script), ss, curl, nikto (solo pruebas no destructivas).
- Ejemplos:

nmap -sV --script=vuln 192.168.56.75

```
(kali@kali)-[~]
$ ping 192.168.56.75
PING 192.168.56.75 (192.168.56.75) 56(84) bytes of data.
64 bytes from 192.168.56.75: icmp_seq=1 ttl=64 time=1.78 ms
64 bytes from 192.168.56.75: icmp_seq=2 ttl=64 time=1.28 ms
64 bytes from 192.168.56.75: icmp_seq=3 ttl=64 time=0.718 ms
64 bytes from 192.168.56.75: icmp_seq=4 ttl=64 time=0.941 ms
64 bytes from 192.168.56.75: icmp_seq=5 ttl=64 time=1.39 ms
64 bytes from 192.168.56.75: icmp_seq=6 ttl=64 time=1.04 ms
^C
— 192.168.56.75 ping statistics —
6 packets transmitted, 6 received, 0% packet loss, time 5037ms
rtt min/avg/max/mdev = 0.718/1.190/1.777/0.341 ms

(kali@kali)-[~]
$ nmap -sV --script=vuln 192.168.56.75
Starting Nmap 7.95 (https://nmap.org) at 2025-10-07 12:27 CEST
Nmap scan report for 192.168.56.75
Host is up (0.00027s latency).
All 1000 scanned ports on 192.168.56.75 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:5F:6D:09 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.45 seconds

(kali@kali)-[~]
$
```

curl -I http://192.168.56.75:80

nikto -host <http://192.168.56.75:80>

```
(kali@kali)-[~]
$ curl -I http://192.168.56.75:80
curl: (7) Failed to connect to 192.168.56.75 port 80 after 0 ms: Could not connect to server

(kali@kali)-[~]
$ nikto -host http://192.168.56.75:80
- Nikto v2.5.0

+ 0 host(s) tested

(kali@kali)-[~]
$
```

**Nota:** no realizar explotación sin autorización.

#### Tareas:

- Comprobar encabezados HTTP, existencia de directorios sensibles, versiones de servicios.
- Anotar hallazgos: servicios sin TLS, versiones obsoletas, páginas de administración accesibles, etc.

## FASE 4 — Análisis de logs y forense básico

**Objetivo:** revisar logs del sistema/servicios para identificar actividad sospechosa.

#### Ubicaciones y herramientas:

- Logs de sistema: /var/log/syslog, /var/log/auth.log.
- Logs de aplicaciones: Squid /var/log/squid/access.log, web /var/log/apache2/access.log.
- Herramientas: grep, ausearch(auditd), journalctl, ELK/Graylog (si disponible).
- Comandos útiles:
- sudo journalctl -u ssh

```
(kali@kali)-[/var/log]
$ sudo journalctl -u ssh
-- No entries --

(kali@kali)-[/var/log]
$
```

**Tareas:**

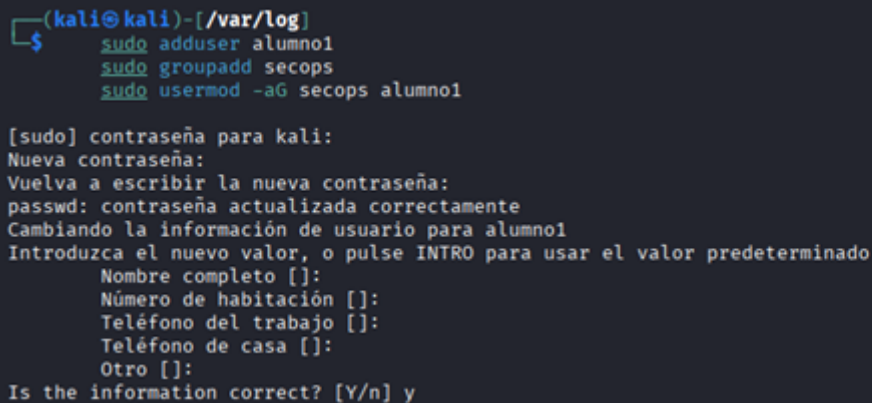
- Identificar intentos fallidos de login, accesos a URLs no autorizadas, timestamps correlacionados con capturas pcap.
  - Generar un timeline básico (CSV) con evento, hora, IP origen, descripción.
- 

**FASE 5 — Gestión de usuarios y permisos**

**Objetivo:** demostrar creación de cuentas, grupos y buenas prácticas y auditar permisos.

**Comandos (ejemplos):**

- Crear usuario y grupo:
- `sudo adduser alumno1`
- `sudo groupadd secops`
- `sudo usermod -aG secops alumno1`



```
(kali㉿kali)-[/var/log]
└─$ sudo adduser alumno1
 sudo groupadd secops
 sudo usermod -aG secops alumno1

[sudo] contraseña para kali:
Nueva contraseña:
Vuelva a escribir la nueva contraseña:
passwd: contraseña actualizada correctamente
Cambiando la información de usuario para alumno1
Introduzca el nuevo valor, o pulse INTRO para usar el valor predeterminado
 Nombre completo []:
 Número de habitación []:
 Teléfono del trabajo []:
 Teléfono de casa []:
 Otro []:
Is the information correct? [Y/n] y
```

- Asignar permisos a directorio:
- `sudo chown root:secops /srv/Seguro`
- `sudo chmod 750 /srv/seguro`

```
(kali@kali)-[/var/log]
$ sudo chown root:secops /srv/seguro
chown: no se puede acceder a '/srv/seguro': No existe el fichero o el directorio
(kali@kali)-[/var/log]
$ cd /srv
(kali@kali)-[/srv]
$ sudo mkdir seguro
(kali@kali)-[/srv]
$ sudo chown root:secops /srv/seguro
(kali@kali)-[/srv]
$ sudo chmod 750 /srv/seguro
(kali@kali)-[/srv]
$
```

- Revisar sudoers (usar visudo):

sudo visudo



```

This file MUST be edited with the 'visudo' command as root.
#
Please consider adding local content in /etc/sudoers.d/ instead of
directly modifying this file.
#
See the man page for details on how to write a sudoers file.
Defaults env_reset
Defaults mail_badpass
Defaults secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
This fixes CUI-DHCP and possibly breaks some versions of dhclient
(Felixclauson, https://bugs.kde.org/view.php?id=380133)
Defaults use_pty

This preserves group settings from user environments of root
equivalent users (group sudo)
Defaults:sudo env_keep += "sudo_group sudo_group_pk_group sudo_group_no_group"

This allows running arbitrary commands, but so does ALL, and it means
different numbers have their choice of editor expected.
Defaults:sudo env_keep += "EDITOR"

Completely harmless preservation of a user preference.
Defaults:sudo env_keep += "XDG_CURRENT_DESKTOP"

While you shouldn't normally run glib as root, you need to with atksuper
Defaults:sudo env_keep += "Glib_Autoshell = Glib_Commandline"

Per-user preferences; root won't have sensible values for them.
Defaults:sudo env_keep += "XMAIL_DIRECTORY_DIRECTORY"

"sudo tcp" or "sudo rsh" should be able to use your SSH agent.
Defaults:sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

Hint for GPG agent
Defaults:sudo env_keep += "GPG_AGENT_INFO"

Most alias specification
User alias specification
Host alias specification
User privilege specification
root ALL=(ALL:ALL) ALL

Allow members of group sudo to execute any command
sudo ALL=(ALL:ALL) ALL

```

```

Allow members of group sudo to execute any command
%sudo ALL=(ALL:ALL) ALL

See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d

```

## Tareas:

- Crear 1 usuario por sub-equipo, asignar grupo y dar acceso restringido a un directorio.
- Documentar comandos y verificar con id usuario y ls -ld.

## FASE 6 — Endurecimiento y medidas de protección (firewall, IDS, mitigaciones)

**Objetivo:** aplicar controles para reducir la superficie de ataque.

### Ejemplos de acciones y comandos:

#### Firewall con nftables (resumen)

- Tabla y cadena básica:

```
nft add table inet filtro
```

```
nft add chain inet filtro input '{ type filter hook input priority 0; policy drop; }'
```

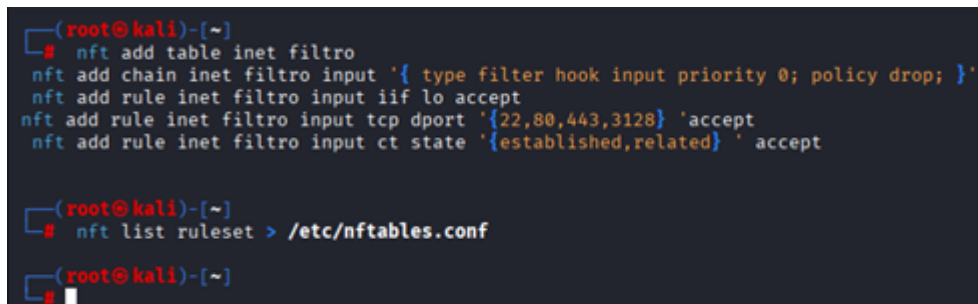
```
nft add rule inet filtro input iif lo accept
```

```
nft add rule inet filtro input tcp dport '{22,80,443,3128}' accept
```

```
nft add rule inet filtro input ct state '{established,related}' accept
```

- Guardar reglas:

```
nft list ruleset > /etc/nftables.conf
```



```
(root@kali)~[~]
nft add table inet filtro
nft add chain inet filtro input '{ type filter hook input priority 0; policy drop; }'
nft add rule inet filtro input iif lo accept
nft add rule inet filtro input tcp dport '{22,80,443,3128}' accept
nft add rule inet filtro input ct state '{established,related}' accept

(root@kali)~[~]
nft list ruleset > /etc/nftables.conf

(root@kali)~[~]
#
```

no me deja guardar la table sin root

#### Firewall con ufw (sencillo)

```
sudo ufw allow 22/tcp
```

```
sudo ufw allow 3128/tcp
```

```
sudo ufw enable
```

```
(kali@kali)-[~]
$ sudo ufw allow 22/tcp
sudo ufw allow 3128/tcp
sudo ufw enable

Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Firewall is active and enabled on system startup

(kali@kali)-[~]
$
```

## HIPS / Hardening

- Habilitar fail2ban para SSH:

sudo apt install fail2ban

sudo systemctl enable --now fail2ban

```
(kali@kali)-[~]
$ sudo apt install fail2ban
Installing:
fail2ban

Installing dependencies:
python3-systemd

Paquetes sugeridos:
naila system-log-daemon monit

Summary:
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 591
Download size: 507 kB
Space needed: 2.461 kB / 32.4 GB available

Continue? [Y/n] y
Des:1 http://http.kali.org/kali kali-rolling/main amd64 python3-systemd amd64 235-1+b6 [40.8 kB]
Des:2 http://kali.download/kali kali-rolling/main amd64 fail2ban all 1.1.0-8 [466 kB]
Descargados 507 kB en 1s (593 kB/s)
Seleccionando el paquete python3-systemd previamente no seleccionado.
(Leyendo la base de datos ... 410781 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../python3-systemd_235-1+b6_amd64.deb ...
Desempaquetando python3-systemd (235-1+b6) ...
Seleccionando el paquete fail2ban previamente no seleccionado.
Preparando para desempaquetar .../fail2ban_1.1.0-8_all.deb ...
Desempaquetando fail2ban (1.1.0-8) ...
Configurando python3-systemd (235-1+b6) ...
Configurando fail2ban (1.1.0-8) ...
update-rc.d: We have no instructions for the fail2ban init script.
update-rc.d: It looks like a network service, we disable it.
fail2ban.service is a disabled or a static unit, not starting it.
Procesando disparadores para kali-menu (2025.3.2) ...
Procesando disparadores para nan-db (2.13.1-1) ...

(kali@kali)-[~]
$ sudo systemctl enable --now fail2ban
Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
Created symlink '/etc/systemd/system/multi-user.target.wants/fail2ban.service' → '/usr/lib/systemd/system/fail2ban.service'.
```

- Auditar con lynis:

sudo apt install lynis

sudo lynis audit system

```
(kali@kali)~$ sudo apt install lynis

Installing:
 lynis

Installing dependencies:
 menu

Paquetes sugeridos:
 apt-listbugs debsecan debsums tripwire sanhain aide menu-l10n

Summary:
 Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 591
 Download size: 629 kB
 Space needed: 3.282 kB / 32,4 GB available

Continue? [5/n] s
Des:1 http://kali.download/kali kali-rolling/main amd64 lynis all 3.1.4-1 [273 kB]
Des:2 http://kali.download/kali kali-rolling/main amd64 menu amd64 2.1.51 [355 kB]
Descargados 629 kB en 1s (1.039 kB/s)
Seleccionando el paquete lynis previamente no seleccionado.
(Leyendo la base de datos ... 417270 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../archives/lynis_3.1.4-1_all.deb ...
Desempaquetando lynis (3.1.4-1) ...
Seleccionando el paquete menu previamente no seleccionado.
Preparando para desempaquetar .../archives/menu_2.1.51_amd64.deb ...
Desempaquetando menu (2.1.51) ...
Configurando lynis (3.1.4-1) ...
Created symlink /etc/systemd/system/timers.target.wants/lynis.timer' + '/usr/lib/systemd/system/lynis.timer'.
lynis.service is a disabled or a static unit, not starting it.
Configurando menu (2.1.51) ...
Procesando disparadores para desktop-file-utils (0.28-1) ...
Procesando disparadores para doc-base (0.11.2) ...
Processing 1 added doc-base file ...
Procesando disparadores para man-db (2.13.1-1) ...
Procesando disparadores para kali-menu (2025.3.2) ...
Procesando disparadores para menu (2.1.51) ...

(kali@kali)~$ sudo lynis audit system

[lynis 3.1.4]

=====
lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.

2007-2024, CISOFY - https://cisofy.com/lynis/
```

```

(kali@kali)~$ sudo lynis audit system

[Lynis 3.1.4]

=====
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.

2007-2024, CISofy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
=====

[+] Initializing program

- Detecting OS ... [DONE]
- Checking profiles ... [DONE]
- Detecting language and localization [es]

Program version: 3.1.4
Operating system: Linux
Operating system name: Kali Linux
Operating system version: Rolling release
Kernel version: 6.12.38+kali
Hardware platform: x86_64
Hostname: kali

Profiles: /etc/lynis/default.prf
Log file: /var/log/lynis.log
Report file: /var/log/lynis-report.dat
Report version: 1.0
Plugin directory: /etc/lynis/plugins

Auditor: [Not Specified]
Language: es
Test category: all
Test group: all

- Program update status ... [SIN ACTUALIZACIÓN]

[+] Herramientas del sistema

- Scanning available tools ...
- Checking system binaries ...

[+] Plugins (fase 1)

Nota: los plugins contienen pruebas más extensivas y toman más tiempo

- Plugin: debian
[

```

```

[+] Herramientas del sistema

- Scanning available tools ...
- Checking system binaries ...

[+] Plugins (fase 1)

Nota: los plugins contienen pruebas más extensivas y toman más tiempo

- Plugin: debian
[
[+] Debian Tests

- Checking for system binaries that are required by Debian Tests ...
- Checking /bin ... [FOUND]
- Checking /sbin ... [FOUND]
- Checking /usr/bin ... [FOUND]
- Checking /usr/sbin ... [FOUND]
- Checking /usr/local/bin ... [FOUND]
- Checking /usr/local/sbin ... [FOUND]
- Authentication:
- PAM (Pluggable Authentication Modules):

WARNING: Test DEB-0001 had a long execution: 11.000509 seconds

- libpam-tmpdir [Not Installed]
- File System Checks:
- DM-Crypt, Cryptsetup & Cryptomount:
- Software:
- apt-listbugs [Not Installed]
- apt-listchanges [Not Installed]
- needrestart [Not Installed]
- fail2ban [Installed with jail.conf]
]

[+] Arranque y servicios

- Service Manager [systemd]
- Checking UEFI boot [DESHABILITADO]
- Checking presence GRUB2 [ENCONTRADO]
- Check running services (systemctl) [RECHO]
 Result: Found 19 running services
- Check enabled services at boot (systemctl) [RECHO]
 Result: Found 19 enabled services
- Check startup files (permissions) [OK]
- Running 'systemd-analyze security'
 Unit name (exposure value) and predicate
- NetworkManager.service (value=6.3) [MEDIO]
- NetworkManager.service (value=7.8) [EXPOSTO]
- accounts-daemon.service (value=5.5) [MEDIO]
- colord.service (value=3.5) [PROTEGIDO]

```

```

- Service Manager [systemd]
- Checking UEFI boot [DESHABILITADO]
- Checking presence GRUB2 [ENCONTRADO]
- Checking for password protection [NINGUNO]
- Check running services (systemctl) [HECHO]
 Result: found 19 running services
- Check enabled services at boot (systemctl) [HECHO]
 Result: found 19 enabled services
- Check startup files (permissions) [OK]
- Running 'systemd-analyze security'
 Unit name (exposure value) and predicate
- ModemManager.service (value=6.3) [MEDIO]
- NetworkManager.service (value=7.8) [EXPUESTO]
- accounts-daemon.service (value=5.5) [MEDIO]
- colord.service (value=3.5) [PROTEGIDO]
- cron.service (value=9.6) [INSEGURO]
- dbus.service (value=9.3) [INSEGURO]
- emergency.service (value=9.5) [INSEGURO]
- fail2ban.service (value=9.6) [INSEGURO]
- getty@tty1.service (value=9.6) [INSEGURO]
- haveged.service (value=3.2) [PROTEGIDO]
- lightdm.service (value=9.6) [INSEGURO]
- lynis.service (value=9.6) [INSEGURO]
- pcsd.service (value=2.1) [PROTEGIDO]
- plymouth-start.service (value=9.5) [INSEGURO]
- polkit.service (value=1.2) [PROTEGIDO]
- rc-local.service (value=9.6) [INSEGURO]
- rescue.service (value=9.5) [INSEGURO]
- rpc-gssd.service (value=9.5) [INSEGURO]
- rpc-statd-notify.service (value=9.5) [INSEGURO]
- rpc-svcgssd.service (value=9.5) [INSEGURO]
- rtkit-daemon.service (value=7.2) [MEDIO]
- smartmontools.service (value=9.6) [INSEGURO]
- ssh.service (value=9.6) [INSEGURO]
- systemd-ask-password-console.service (value=9.4) [INSEGURO]
- systemd-ask-password-plymouth.service (value=9.5) [INSEGURO]
- systemd-ask-password-wall.service (value=9.4) [INSEGURO]
- systemd-bsod.service (value=9.5) [INSEGURO]
- systemd-hostnamed.service (value=1.7) [PROTEGIDO]
- systemd-initctl.service (value=9.4) [INSEGURO]
- systemd-journald.service (value=4.9) [PROTEGIDO]
- systemd-logind.service (value=2.8) [PROTEGIDO]
- systemd-networkd.service (value=2.9) [PROTEGIDO]
- systemd-rfkill.service (value=9.4) [INSEGURO]
- systemd-timesyncd.service (value=2.1) [PROTEGIDO]
- systemd-udev.service (value=7.1) [MEDIO]
- udisks2.service (value=9.6) [INSEGURO]
- upower.service (value=2.4) [PROTEGIDO]
- user@1000.service (value=9.4) [INSEGURO]
- virtualbox-guest-utils.service (value=9.6) [INSEGURO]

```

```
[+] Kernel
- Checking default runlevel [runlevel 5]
- Checking CPU support (NX/PAE) [ENCONTRADO]
 CPU support: PAE and/or NoeXecute supported [HECHO]
- Checking kernel version and release [HECHO]
- Checking kernel type [HECHO]
- Checking loaded kernel modules [HECHO]
 Found 112 active modules
- Checking Linux kernel configuration file [ENCONTRADO]
- Checking default I/O kernel scheduler [NO ENCONTRADO]
- Checking for available kernel update [OK]
- Checking core dumps configuration
 - configuration in systemd conf files [POR DEFECTO]
 - configuration in /etc/profile [POR DEFECTO]
 - 'hard' configuration in /etc/security/limits.conf [ENABLED]
 - 'soft' configuration in /etc/security/limits.conf [DISABLED]
- Checking setuid core dumps configuration [DESHABILITADO]
- Check if reboot is needed [NO]

[+] Memoria y procesos
- Checking /proc/meminfo [ENCONTRADO]
- Searching for dead/zombie processes [NO ENCONTRADO]
- Searching for IO waiting processes [NO ENCONTRADO]
- Search prelink tooling [NO ENCONTRADO]

[+] Usuarios, grupos y autenticación
- Administrator accounts [OK]
- Unique UIDs [OK]
- Consistency of group files (grpck) [OK]
- Unique group IDs [OK]
- Unique group names [OK]
- Password file consistency [OK]
- Password hashing methods [OK]
- Checking password hashing rounds [DESHABILITADO]
- Query system users (non daemons) [HECHO]
- NIS+ authentication support [NO HABILITADO]
- NIS authentication support [NO HABILITADO]
- Sudoers file(s) [ENCONTRADO]
 - Permissions for directory: /etc/sudoers.d [PELIGRO]
 - Permissions for: /etc/sudoers [OK]
 - Permissions for: /etc/sudoers.d/kali-grant-root [OK]
 - Permissions for: /etc/sudoers.d/ospd-openvas [OK]
 - Permissions for: /etc/sudoers.d/README [OK]
- PAM password strength tools [SUGERENCIA]
- PAM configuration files (pam.conf) [ENCONTRADO]
- PAM configuration files (pam.d) [ENCONTRADO]
- PAM modules [ENCONTRADO]
- LDAP module in PAM [NO ENCONTRADO]
- Accounts without expire date [SUGERENCIA]
```



|                                                       |                   |
|-------------------------------------------------------|-------------------|
| [+] Usuarios, grupos y autenticación                  |                   |
| - Administrator accounts                              | [ OK ]            |
| - Unique UIDs                                         | [ OK ]            |
| - Consistency of group files (grpck)                  | [ OK ]            |
| - Unique group IDs                                    | [ OK ]            |
| - Unique group names                                  | [ OK ]            |
| - Password file consistency                           | [ OK ]            |
| - Password hashing methods                            | [ DESHABILITADO ] |
| - Checking password hashing rounds                    | [ HECHO ]         |
| - Query system users (non daemons)                    | [ NO HABILITADO ] |
| - NIS+ authentication support                         | [ NO HABILITADO ] |
| - NIS authentication support                          | [ ENCONTRADO ]    |
| - Sudoers file(s)                                     | [ PELIGRO ]       |
| - Permissions for directory: /etc/sudoers.d           | [ OK ]            |
| - Permissions for: /etc/sudoers                       | [ OK ]            |
| - Permissions for: /etc/sudoers.d/kali-grant-root     | [ OK ]            |
| - Permissions for: /etc/sudoers.d/osspd-openvas       | [ OK ]            |
| - Permissions for: /etc/sudoers.d/README              | [ OK ]            |
| - PAM password strength tools                         | [ SUGERENCIA ]    |
| - PAM configuration files (pam.conf)                  | [ ENCONTRADO ]    |
| - PAM configuration files (pam.d)                     | [ ENCONTRADO ]    |
| - PAM modules                                         | [ ENCONTRADO ]    |
| - LDAP module in PAM                                  | [ NO ENCONTRADO ] |
| - Accounts without expire date                        | [ SUGERENCIA ]    |
| - Accounts without password                           | [ OK ]            |
| - Locked accounts                                     | [ OK ]            |
| - Checking user password aging (minimum)              | [ DESHABILITADO ] |
| - User password aging (maximum)                       | [ DESHABILITADO ] |
| - Checking expired passwords                          | [ OK ]            |
| - Checking linux single user mode authentication      | [ OK ]            |
| - Determining default umask                           |                   |
| - umask (/etc/profile)                                | [ NO ENCONTRADO ] |
| - umask (/etc/login.defs)                             | [ SUGERENCIA ]    |
| - LDAP authentication support                         | [ NO HABILITADO ] |
| - Logging failed login attempts                       | [ DESHABILITADO ] |
| [+] Kerberos                                          |                   |
| - Check for Kerberos KDC and principals               | [ NO ENCONTRADO ] |
| [+] Shells                                            |                   |
| - Checking shells from /etc/shells                    |                   |
| Result: found 13 shells (valid shells: 13).           |                   |
| - Session timeout settings/tools                      | [ NINGUNO ]       |
| - Checking default umask values                       |                   |
| - Checking default umask in /etc/bash.bashrc          | [ NINGUNO ]       |
| - Checking default umask in /etc/profile              | [ NINGUNO ]       |
| [+] Sistemas de ficheros                              |                   |
| - Permissions of home directories                     | [ PELIGRO ]       |
| - Ownership of home directories                       | [ OK ]            |
| - Checking shell history files                        | [ OK ]            |
| [+] Bastionado del kernel                             |                   |
| - Comparing sysctl key pairs with scan profile        |                   |
| - dev.tty.ldisc_autoload (exp: 0)                     | [ DIFERENTE ]     |
| - fs.protected_fifos (exp: 2)                         | [ DIFERENTE ]     |
| - fs.protected_hardlinks (exp: 1)                     | [ OK ]            |
| - fs.protected_regular (exp: 2)                       | [ OK ]            |
| - fs.protected_symlinks (exp: 1)                      | [ OK ]            |
| - fs.suid_dumpable (exp: 0)                           | [ OK ]            |
| - kernel.core_uses_pid (exp: 1)                       | [ OK ]            |
| - kernel.ctrl-alt-del (exp: 0)                        | [ OK ]            |
| - kernel.dmesg_restrict (exp: 1)                      | [ DIFERENTE ]     |
| - kernel.kptr_restrict (exp: 2)                       | [ DIFERENTE ]     |
| - kernel.modules_disabled (exp: 1)                    | [ DIFERENTE ]     |
| - kernel.perf_event_paranoid (exp: 2 3 4)             | [ OK ]            |
| - kernel.randomize_va_space (exp: 2)                  | [ OK ]            |
| - kernel.sysrq (exp: 0)                               | [ DIFERENTE ]     |
| - kernel.unprivileged_bpf_disabled (exp: 1)           | [ DIFERENTE ]     |
| - kernel.yama.ptrace_scope (exp: 1 2 3)               | [ DIFERENTE ]     |
| - net.core.bpf_jit_harden (exp: 2)                    | [ DIFERENTE ]     |
| - net.ipv4.conf.all.accept_redirects (exp: 0)         | [ OK ]            |
| - net.ipv4.conf.all.accept_source_route (exp: 0)      | [ OK ]            |
| - net.ipv4.conf.all.bootp_relay (exp: 0)              | [ OK ]            |
| - net.ipv4.conf.all.forwarding (exp: 0)               | [ OK ]            |
| - net.ipv4.conf.all.log_martians (exp: 1)             | [ DIFERENTE ]     |
| - net.ipv4.conf.all.mc_forwarding (exp: 0)            | [ OK ]            |
| - net.ipv4.conf.all.proxy_arp (exp: 0)                | [ OK ]            |
| - net.ipv4.conf.all.rp_filter (exp: 1)                | [ DIFERENTE ]     |
| - net.ipv4.conf.all.send_redirects (exp: 0)           | [ DIFERENTE ]     |
| - net.ipv4.conf.default.accept_redirects (exp: 0)     | [ OK ]            |
| - net.ipv4.conf.default.accept_source_route (exp: 0)  | [ OK ]            |
| - net.ipv4.conf.default.log_martians (exp: 1)         | [ DIFERENTE ]     |
| - net.ipv4.icmp_echo_ignore_broadcasts (exp: 1)       | [ OK ]            |
| - net.ipv4.icmp_ignore_bogus_error_responses (exp: 1) | [ OK ]            |
| - net.ipv4.tcp_syncookies (exp: 1)                    | [ OK ]            |
| - net.ipv4.tcp_timestamps (exp: 0 1)                  | [ OK ]            |
| - net.ipv6.conf.all.accept_redirects (exp: 0)         | [ OK ]            |
| - net.ipv6.conf.all.accept_source_route (exp: 0)      | [ OK ]            |
| - net.ipv6.conf.default.accept_redirects (exp: 0)     | [ OK ]            |
| - net.ipv6.conf.default.accept_source_route (exp: 0)  | [ OK ]            |
| [+] Bastionado                                        |                   |
| - Installed compiler(s)                               | [ ENCONTRADO ]    |
| - Installed malware scanner                           | [ NO ENCONTRADO ] |
| - Non-native binary formats                           | [ ENCONTRADO ]    |
| [+] Pruebas personalizadas                            |                   |
| - Running custom tests...                             | [ NINGUNO ]       |



**Tareas:**

- Aplicar reglas mínimas de firewall que permitan solo lo necesario.
- Activar fail2ban y demostrar que bloquea after X intentos (simulación).
- Documentar configuración y justificar decisiones.

**FASE 7 — Informe final y presentación****Entregables (mínimos):**

1. report/Informe\_EquipoX.pdf con:
  - Alcance y reglas del laboratorio.
  - Resumen de hallazgos (hosts, puertos, servicios).
  - Evidencias (pcap snippets, logs extractados, capturas).
  - Riesgos priorizados y medidas de remediación (con comandos).
  - Cronograma y roles.
2. Carpeta pcaps/, logs/, scripts/.
3. Diapositiva para presentación

**Herramientas del módulo «Seguridad informática» (recordatorio para la práctica)**

- Recon / Scanning: nmap, netdiscover, arp-scan, masscan (con precaución).
- Captura / Análisis de tráfico: tcpdump, tshark, Wireshark, iftop, nethogs.
- Packet crafting / análisis: scapy, pyshark.
- Logs / SIEM básico: journalctl, auditd, rsyslog, ELK (Elasticsearch + Logstash + Kibana) o Splunk (si disponible).
- Forense / integridad: sleuthkit, autopsy, chkrootkit, rkhunter.
- Hardening / auditoría: ufw, nftables, iptables (legacy), fail2ban, lynis.
- Proxy / Web: squid, apache2, nginx, nikto.
- Gestión de usuarios/permisos: adduser, groupadd, usermod, chmod, chown, visudo.

- Scripting y automatización: Bash, Python (pandas, scikit-learn opcional), Streamlit (visual).
  - Documentación / versiones: git.
- 

### **Ejercicios concretos para entregar (lista rápida)**

1. recon/hosts.csv + nmap output.
  2. pcaps/pcap1.pcap con explicación de 3 flujos.
  3. logs/logs\_issue.txt con 3 hallazgos importantes extraídos de logs.
  4. hardening/ con nftables.conf y pasos ejecutados (comandos).
  5. report/Informe\_EquipoX.pdf y 5–8 diapositivas resumen.
- 

Optativo quien quiera realizar una exposición en grupo del trabajo realizado o al menos explicar al resto de compañeros lo que han realizado.

## FASE 1 — Reconocimiento y mapeo de red

**Objetivo:** identificar hosts activos, puertos y servicios.

### Herramientas y comandos (ejemplos):

- Descubrimiento de hosts:
  - arp-scan (si está permitido):
  - sudo arp-scan -l
  - netdiscover:
  - sudo netdiscover -r 192.168.56.0/24
- Escaneo de puertos con nmap:
  - Scan rápido de puertos y servicios:
  - nmap -sS -sV -O -Pn 192.168.56.0/24
  - Escaneo de puertos específicos:
  - nmap -p 22,80,443,3128 192.168.56.101
- Resultado esperado: lista de hosts, puertos abiertos, servicios y versiones.

### Tareas del equipo:

- Recon/Red hace mapeo y sube hosts.csv con IP, hostname, puertos y servicios detectados.
- Indicar servicios críticos (SSH, HTTP, RDP, DB, proxy).

Preparamos carpetas en Kali:

```
(kali㉿kali)-[~]
$ mkdir -p ~/lab/recon ~/lab/recon/nmap_outputs

(kali㉿kali)-[~]
$ cd ~/lab/recon

(kali㉿kali)-[~/lab/recon]
```

Descubrimos hosts activos:

```
Session Acciones Editar Vista Ayuda
GNU nano 8.6
192.168.56.17
192.168.56.22
192.168.56.40
192.168.56.75
192.168.56.80
192.168.56.81
192.168.56.82
192.168.56.70
```

Hacemos un escaneo rápido de puertos:

```
(kali㉿kali)-[~]
$ sudo nmap -sS -T4 -p 1-1024 --open -oG recon_quick.gnmap 192.168.56.0/24
Starting Nmap 7.95 (https://nmap.org) at 2025-10-03 13:51 CEST
Nmap scan report for 192.168.56.81
Host is up (0.00034s latency).
Not shown: 1023 closed tcp ports (reset)
PORT STATE SERVICE
22/tcp open ssh
MAC Address: 08:00:27:45:BC:2B (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 256 IP addresses (9 hosts up) scanned in 2.24 seconds
```

Ahora un escaneo de puertos específicos:

```
(kali㉿kali)-[~]
$ nmap -p 22,80,443,3128 192.168.56.101
Starting Nmap 7.95 (https://nmap.org) at 2025-10-03 13:53 CEST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.46 seconds
```

## FASE 2 — Análisis de tráfico y paquetes

**Objetivo:** capturar tráfico relevante y analizar patrones/anomalías.

**Herramientas y comandos:**

- Captura de paquetes:
  - tcpdump para capturar en interfaz:
  - `sudo tcpdump -i eth0 -w /home/kali/pcaps/scan_capture.pcap`
  - Wireshark

```
(kali㉿kali)-[~]
$ sudo tcpdump -i eth1 -w /home/kali/pcaps/scan_capture.pcap
tcpdump: listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C4994 packets captured
5031 packets received by filter
0 packets dropped by kernel
```

- Monitoreo en vivo:
  - iftop (consumo por conx):
  - sudo iftop -i eth0
  - nethogs (por proceso)

| Session         | Acciones | Editar | Vista | Ayuda | 12,5Kb          | 25,0Kb          | 37,5Kb          | 50,0Kb          | 62,5Kb          |
|-----------------|----------|--------|-------|-------|-----------------|-----------------|-----------------|-----------------|-----------------|
| 155.255.255.255 |          |        |       |       | ⇒ 192.168.0.66  | ⇒ 192.168.0.66  | ⇒ 192.168.0.66  | ⇒ 192.168.0.66  | ⇒ 192.168.0.66  |
| 192.168.1.255   |          |        |       |       | ⇒ 192.168.0.124 | ⇒ 192.168.0.124 | ⇒ 192.168.0.124 | ⇒ 192.168.0.124 | ⇒ 192.168.0.124 |
| 192.168.1.255   |          |        |       |       | ⇒ 192.168.0.123 | ⇒ 192.168.0.123 | ⇒ 192.168.0.123 | ⇒ 192.168.0.123 | ⇒ 192.168.0.123 |
| 192.168.1.255   |          |        |       |       | ⇒ 192.168.0.66  | ⇒ 192.168.0.66  | ⇒ 192.168.0.66  | ⇒ 192.168.0.66  | ⇒ 192.168.0.66  |
| 192.168.1.255   |          |        |       |       | ⇒ 192.168.0.88  | ⇒ 192.168.0.88  | ⇒ 192.168.0.88  | ⇒ 192.168.0.88  | ⇒ 192.168.0.88  |
| 192.168.1.255   |          |        |       |       | ⇒ 192.168.0.118 | ⇒ 192.168.0.118 | ⇒ 192.168.0.118 | ⇒ 192.168.0.118 | ⇒ 192.168.0.118 |
| 192.168.1.255   |          |        |       |       | ⇒ 192.168.0.100 | ⇒ 192.168.0.100 | ⇒ 192.168.0.100 | ⇒ 192.168.0.100 | ⇒ 192.168.0.100 |
| 255.255.255.255 |          |        |       |       | ⇒ 0.0.0.0       | ⇒ 0.0.0.0       | ⇒ 0.0.0.0       | ⇒ 0.0.0.0       | ⇒ 0.0.0.0       |

- Análisis de .pcap:
  - Abrir pcap con **Wireshark** (GUI) o tshark:
  - tshark -r scan\_capture.pcap -q -z io,phs
- Extracción de flujos/estadísticas con pyshark / scripts Python (opcional).

```
(kali@kali)-[~]
$ tshark -r scan_capture.pcap -q -z io,phs

Protocol Hierarchy Statistics
Filter:

json frames:1 bytes:5
data-text-lines frames:1 bytes:5
```

### Tareas del equipo:

- Capturar tráfico durante un periodo de actividad (p. ej. navegación desde cliente a través de proxy).
- Extraer 3 flujos relevantes y explicar (IP origen/destino, puertos, bytes, flags).
- Detectar posibles anomalías (tráfico inesperado, puertos inusuales).

## FASE 3 — Enumeración de servicios y pruebas seguras

**Objetivo:** verificar configuraciones visibles (vulnerabilidades de servicio *sin explotar*).

### Herramientas:

- nmap scripts (--script), ss, curl, nikto (solo pruebas no destructivas).
- Ejemplos:
- nmap -sV --script=vuln 192.168.56.101

```
(kali㉿kali)-[~]
$ nmap -sV --script=vuln 192.168.56.101
Starting Nmap 7.95 (https://nmap.org) at 2025-10-03 14:07 CEST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 11.69 seconds
```

- curl -I <http://192.168.56.101:80>

```
(kali㉿kali)-[~]
$ curl -I http://192.168.56.101:80
curl: (7) Failed to connect to 192.168.56.101 port 80 after 3058 ms: Could not connect to server
```

- nikto -host <http://192.168.56.101:80>

```
(kali㉿kali)-[~]
$ nikto -host http://192.168.56.101:80

- Nikto v2.5.0

+ 0 host(s) tested
```

**Nota:** no realizar explotación sin autorización.

#### Tareas:

- Comprobar encabezados HTTP, existencia de directorios sensibles, versiones de servicios.
- Anotar hallazgos: servicios sin TLS, versiones obsoletas, páginas de administración accesibles, etc.

---

## FASE 4 — Análisis de logs y forense básico

**Objetivo:** revisar logs del sistema/servicios para identificar actividad sospechosa.

#### Ubicaciones y herramientas:

- Logs de sistema: /var/log/syslog, /var/log/auth.log.
- Logs de aplicaciones: Squid /var/log/squid/access.log, web /var/log/apache2/access.log.
- Herramientas: grep, ausearch(auditd), journalctl, ELK/Graylog (si disponible).
- Comandos útiles:
- sudo tail -n 200 /var/log/auth.log

```
(kali㉿kali)-[/var/log]
$ sudo tail -n 200 /var/log/auth.log
```

a

```
(kali㉿kali)-[/var/log]
$
```

- `sudo grep "Failed password" /var/log/auth.log | wc -l`

```
(kali㉿kali)-[/var/log]
$ sudo grep "Failed password" /var/log/auth.log | wc -l
0
```

- `sudo journalctl -u ssh`

```
(kali㉿kali)-[/var/log]
$ sudo journalctl -u ssh
oct 03 13:38:52 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
oct 03 13:38:52 kali sshd[5657]: Server listening on 0.0.0.0 port 22.
oct 03 13:38:52 kali sshd[5657]: Server listening on :: port 22.
oct 03 13:38:52 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
oct 03 13:39:11 kali sshd-session[5804]: Accepted password for kali from 192.168.0.42 port 50784 ssh2
oct 03 13:39:11 kali sshd-session[5804]: pam_unix(sshd:session): session opened for user kali(uid=1000) by kali(uid=0)
oct 03 13:41:56 kali systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
oct 03 13:41:56 kali systemd[1]: ssh.service: Deactivated successfully.
oct 03 13:41:56 kali systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
```

### Tareas:

- Identificar intentos fallidos de login, accesos a URLs no autorizadas, timestamps correlacionados con capturas pcap.
- Generar un timeline básico (CSV) con evento, hora, IP origen, descripción.

---

## FASE 5 — Gestión de usuarios y permisos

**Objetivo:** demostrar creación de cuentas, grupos y buenas prácticas y auditar permisos.

### Comandos (ejemplos):

- Crear usuario y grupo:
- `sudo adduser alumno1`

```
(kali㉿kali)-[/var/log]
$ sudo adduser alumno1

Nueva contraseña:
Vuelva a escribir la nueva contraseña:
passwd: contraseña actualizada correctamente
Cambiando la información de usuario para alumno1
Introduzca el nuevo valor, o pulse INTRO para usar el valor predeterminado
 Nombre completo []:
 Número de habitación []:
 Teléfono del trabajo []:
 Teléfono de casa []:
 Otro []:
Is the information correct? [Y/n] y
```

- sudo groupadd secops

```
(kali㉿kali)-[/var/log]
$ sudo groupadd secops
```

- sudo usermod -aG secops alumno1

```
(kali㉿kali)-[/var/log]
$ sudo usermod -aG secops alumno1
```

- Asignar permisos a directorio:
- sudo chown root:secops /srv/seguro

```
(kali㉿kali)-[/srv]
$ sudo chown root:secops /srv/seguro
```

- sudo chmod 750 /srv/seguro

```
(kali㉿kali)-[/srv]
$ sudo chmod 750 /srv/seguro
```

- Revisar sudoers (usar visudo):
- sudo visudo

```
User privilege specification
root ALL=(ALL:ALL) ALL

Allow members of group sudo to execute any command
%sudo ALL=(ALL:ALL) ALL

See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d
```



**Tareas:**

- Crear 1 usuario por sub-equipo, asignar grupo y dar acceso restringido a un directorio.
- Documentar comandos y verificar con id usuario y ls -ld.

**FASE 6 — Endurecimiento y medidas de protección (firewall, IDS, mitigaciones)**

**Objetivo:** aplicar controles para reducir la superficie de ataque.

**Ejemplos de acciones y comandos:****Firewall con nftables (resumen)**

- Tabla y cadena básica:
- `sudo nft add table inet filtro`

```
(kali㉿kali)-[/srv]
$ sudo nft add table inet filtro
```

- `sudo nft add chain inet filtro input { type filter hook input priority 0 \; policy drop \; }`

```
(kali㉿kali)-[/]
$ sudo nft add chain inet filtro input '{ type filter hook input priority 0; policy drop; }'
(kali㉿kali)-[/]
$
```

- `sudo nft add rule inet filtro input iif lo accept`

```
(kali㉿kali)-[/]
$ sudo nft add rule inet filtro input iif lo accept
(kali㉿kali)-[/]
$
```

- `sudo nft add rule inet filtro input tcp dport {22,80,443,3128} accept`

```
(kali㉿kali)-[/]
$ sudo nft add rule inet filtro input tcp dport '{22,80,443,3128}' accept
(kali㉿kali)-[/]
$
```

- `sudo nft add rule inet filtro input ct state {established,related} accept`

```
(kali㉿kali)-[/]
$ sudo nft add rule inet filtro input ct state '{established,related}' accept
(kali㉿kali)-[/]
$
```

- Guardar reglas:
- `sudo nft list ruleset > /etc/nftables.conf`

```
(kali㉿kali)-[/]
$ sudo nft list ruleset > /etc/nftables.conf
zsh: permiso denegado: /etc/nftables.conf
(kali㉿kali)-[/]
$ sudo su
(root㉿kali)-[/]
sudo nft list ruleset > /etc/nftables.conf
(root㉿kali)-[/]
#
```

### Firewall con ufw (sencillo)

`sudo ufw allow 22/tcp`

```
(kali㉿kali)-[/]
$ sudo ufw allow 22/tcp
Rules updated
Rules updated (v6)
(kali㉿kali)-[/]
$
```

`sudo ufw allow 3128/tcp`

```
(kali㉿kali)-[/]
$ sudo ufw allow 3128/tcp
Rules updated
Rules updated (v6)
```

`sudo ufw enable`

```
(kali@kali)-[/]
$ sudo ufw enable
Firewall is active and enabled on system startup

(kali@kali)-[/]
$
```

## HIPS / Hardening

- Habilitar fail2ban para SSH:
- `sudo apt install fail2ban`

```
(kali@kali)-[/]
$ sudo apt install fail2ban

Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios.
anass-common libjs-jquery-ui libmongocrypt0 librav1e0.7 libudfread0 libyelp0 python3-kismetcapturebtgeiger python3-kismetcapturebtldsb samba-ad-dc
libbluray2 libjs-underscore libplacebo349 libtheoradec1 libx264-164 python3-bluepy python3-kismetcapturefreaklabszigbee python3-kismetcapturetlamr samba-ad-provision
libbson-1.0-0t64 libmongoc-1.0-0t64 libportmidi0 libtheoraenc1 libxml2 python3-gpg python3-kismetcapturertl433 python3-protobuf samba-dsdb-modules
Utilice «sudo apt autoremove» para eliminarlos.

Installing:
fail2ban

Installing dependencies:
python3-systemd

Paquetes sugeridos:
mailx system-log-daemon monit

Summary:
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 3
Download size: 507 kB
Space needed: 2.461 kB / 5.895 MB available

Continue? [S/n] s
Des:1 http://http.kali.org/kali kali-rolling/main amd64 python3-systemd amd64 235-1+b6 [40,8 kB]
Des:2 http://mirror.es.cdn-perfprod.com/kali kali-rolling/main amd64 fail2ban all 1.1.0-8 [466 kB]
Descargados 507 kB en 12s (41,4 kB/s)
Seleccionando el paquete python3-systemd previamente no seleccionado.
(Leyendo la base de datos ... 435267 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../python3-systemd_235-1+b6_amd64.deb ...
Desempaquetando python3-systemd (235-1+b6) ...
Seleccionando el paquete fail2ban previamente no seleccionado.
Preparando para desempaquetar .../fail2ban_1.1.0-8_all.deb ...
Desempaquetando fail2ban (1.1.0-8) ...
Configurando python3-systemd (235-1+b6) ...
Configurando fail2ban (1.1.0-8) ...
update-rc.d: We have no instructions for the fail2ban init script.
update-rc.d: It looks like a network service, we disable it.
fail2ban.service is a disabled or a static unit, not starting it.
Procesando disparadores para kali-menu (2025.4.1) ...
Procesando disparadores para man-db (2.13.1-1) ...
```

- `sudo systemctl enable --now fail2ban`

```
(kali@kali)-[/]
$ sudo systemctl enable --now fail2ban
Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
Created symlink '/etc/systemd/system/multi-user.target.wants/fail2ban.service' → '/usr/lib/systemd/system/fail2ban.service'.
```

- Auditar con lynis:
- `sudo apt install lynis`

```

(kali@kali) [/]
$ sudo apt install lynis
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios.
amass-common libjs-jquery-ui libmongocrypt0 libavie0.7 libudfread0 libyelp0 python3-kismetcapturebtgeiger python3-kismetcaptureertladsb samba-ad-dc
libbluray2 libjs-underscore libplacebo349 libtheoradec1 libx264-164 python3-bluepy python3-kismetcapturefreaklabszigbee python3-kismetcaptureertlamr samba-ad-provision
libbson-1.0-0t64 libmongoc-1.0-0t64 libportmidi0 libtheoraenc1 libxml2 python3-gpg python3-kismetcaptureertl433 python3-protobuf samba-dsdb-modules
Utilice «sudo apt autoremove» para eliminarlos.

Installing:
lynis

Installing dependencies:
menu

Paquetes sugeridos:
apt-listbugs debsecan debsums tripwire samhain aide menu-l10n

Summary:
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 3
Download size: 629 kB
Space needed: 3.202 kB / 5.890 MB available

Continue? [S/n] s
Des:1 http://kali.download/kali kali-rolling/main amd64 lynis all 3.1.4-1 [273 kB]
Des:2 http://kali.download/kali kali-rolling/main amd64 menu amd64 2.1.51 [355 kB]
Descargados 629 kB en 1s (1.160 kB/s)
Seleccionando el paquete lynis previamente no seleccionado.
(Leyendo la base de datos ... 435756 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../archives/lynis_3.1.4-1_all.deb ...
Desempaquetando lynis (3.1.4-1) ...
Seleccionando el paquete menu previamente no seleccionado.
Preparando para desempaquetar .../archives/menu_2.1.51_amd64.deb ...
Desempaquetando menu (2.1.51) ...
Configurando lynis (3.1.4-1) ...
Created symlink '/etc/systemd/system/timers.target.wants/lynis.timer' → '/usr/lib/systemd/system/lynis.timer'.
lynis.service is a disabled or a static unit, not starting it.
Configurando menu (2.1.51) ...
Procesando disparadores para desktop-file-utils (0.28-1) ...
Procesando disparadores para doc-base (0.11.2) ...
Processing 1 added doc-base file...
Procesando disparadores para man-db (2.13.1-1) ...
Procesando disparadores para kali-menu (2025.4.1) ...
Procesando disparadores para menu (2.1.51) ...

```

- sudo lynis audit system

### [+] Initializing program

```

- Detecting OS ... [DONE]
- Checking profiles ... [DONE]
- Detecting language and localization [es]

Program version: 3.1.4
Operating system: Linux
Operating system name: Kali Linux
Operating system version: Rolling release
Kernel version: 6.12.38+kali
Hardware platform: x86_64
Hostname: kali

Profiles: /etc/lynis/default.prf
Log file: /var/log/lynis.log
Report file: /var/log/lynis-report.dat
Report version: 1.0
Plugin directory: /etc/lynis/plugins

Auditor: [Not Specified]
Language: es
Test category: all
Test group: all

```

[+] **Debian Tests**

- Checking for system binaries that are required by Debian Tests...
  - Checking /bin... [ FOUND ]
  - Checking /sbin... [ FOUND ]
  - Checking /usr/bin... [ FOUND ]
  - Checking /usr/sbin... [ FOUND ]
  - Checking /usr/local/bin... [ FOUND ]
  - Checking /usr/local/sbin... [ FOUND ]
- Authentication:
  - PAM (Pluggable Authentication Modules):
    - libpam-tmpdir [ Not Installed ]
- File System Checks:
  - DM-Crypt, Cryptsetup & Cryptmount:
- Software:
  - apt-listbugs [ Not Installed ]
  - apt-listchanges [ Not Installed ]
  - needrestart [ Not Installed ]
  - fail2ban [ Installed with jail.conf ]

[+] **Arranque y servicios**

- Service Manager [ systemd ]
- Checking UEFI boot [ DESHABILITADO ]
- Checking presence GRUB2 [ ENCONTRADO ]
  - Checking for password protection [ NINGUNO ]
- Check running services (systemctl) [ HECHO ]
  - Result: found 19 running services
- Check enabled services at boot (systemctl) [ HECHO ]
  - Result: found 21 enabled services
- Check startup files (permissions) [ OK ]
- Running 'systemd-analyze security'
  - Unit name (exposure value) and predicate
- ModemManager.service (value=6.3) [ MEDIO ]
- NetworkManager.service (value=7.8) [ EXPUESTO ]
- accounts-daemon.service (value=5.5) [ MEDIO ]
- colord.service (value=3.5) [ PROTEGIDO ]
- cron.service (value=9.6) [ INSEGURO ]
- dbus.service (value=9.3) [ INSEGURO ]
- emergency.service (value=9.5) [ INSEGURO ]
- fail2ban.service (value=9.6) [ INSEGURO ]
- getty@tty1.service (value=9.6) [ INSEGURO ]
- haveged.service (value=3.2) [ PROTEGIDO ]
- lightdm.service (value=9.6) [ INSEGURO ]
- lynis.service (value=9.6) [ INSEGURO ]
- pcscd.service (value=2.1) [ PROTEGIDO ]
- plymouth-start.service (value=9.5) [ INSEGURO ]

```

Lynis security scan details:

Hardening index : 65 [#####]
Tests performed : 273
Plugins enabled : 1

Components:
- Firewall [V]
- Malware scanner [X]

Scan mode:
Normal [V] Forensics [] Integration [] Pentest []

Lynis modules:
- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

```

#### Tareas:

- Aplicar reglas mínimas de firewall que permitan solo lo necesario.
- Activar fail2ban y demostrar que bloquea after X intentos (simulación).
- Documentar configuración y justificar decisiones.

## FASE 7 — Informe final y presentación

### Entregables (mínimos):

1. report/Informe\_EquipoX.pdf con:
  - Alcance y reglas del laboratorio.
  - Resumen de hallazgos (hosts, puertos, servicios).
  - Evidencias (pcap snippets, logs extractados, capturas).
  - Riesgos priorizados y medidas de remediación (con comandos).
  - Cronograma y roles.
2. Carpeta pcaps/, logs/, scripts/.
3. Diapositiva para presentación

### Herramientas del módulo «Seguridad informática» (recordatorio para la práctica)

- Recon / Scanning: nmap, netdiscover, arp-scan, masscan (con precaución).
- Captura / Análisis de tráfico: tcpdump, tshark, Wireshark, iftop, nethogs.
- Packet crafting / análisis: scapy, pyshark.
- Logs / SIEM básico: journalctl, auditd, rsyslog, ELK (Elasticsearch + Logstash + Kibana) o Splunk (si disponible).
- Forense / integridad: sleuthkit, autopsy, chkrootkit, rkhunter.

- Hardening / auditoría: ufw, nftables, iptables (legacy), fail2ban, lynis.
  - Proxy / Web: squid, apache2, nginx, nikto.
  - Gestión de usuarios/permisos: adduser, groupadd, usermod, chmod, chown, visudo.
  - Scripting y automatización: Bash, Python (pandas, scikit-learn opcional), Streamlit (visual).
  - Documentación / versiones: git.
- 

### **Ejercicios concretos para entregar (lista rápida)**

1. recon/hosts.csv + nmap output.
  2. pcaps/pcap1.pcap con explicación de 3 flujos.
  3. logs/logs\_issue.txt con 3 hallazgos importantes extraídos de logs.
  4. hardening/ con nftables.conf y pasos ejecutados (comandos).
  5. report/Informe\_EquipoX.pdf y 5–8 diapositivas resumen.
- 

Optativo quien quiera realizar una exposición en grupo del trabajo realizado o al menos explicar al resto de compañeros lo que han realizado.

**FASE 0 —**

- Definir ordenadores/segmentos IP

| Host   | IP            | Nombre  |
|--------|---------------|---------|
| Kali 1 | 192.168.56.70 | Gastón  |
| Kali 2 | 192.168.56.75 | Álvaro  |
| Kali 3 | 192.168.56.40 | Daniela |
| Kali 4 | 192.168.56.17 | Dani    |
| Kali 5 | 192.168.56.22 | Rafa    |

- Documentar alcance y reglas (qué está permitido).
- Crear repositorio con estructura: recon/, pcaps/, logs/, report/.

| > Este equipo > Documentos > GitHub > Repositorio-Grupo |                    |                       |                     |        |
|---------------------------------------------------------|--------------------|-----------------------|---------------------|--------|
|                                                         | Nombre             | Fecha de modificación | Tipo                | Tamaño |
| ✦                                                       | Ejercicio ficheros | 03/10/2025 8:33       | Carpeta de archivos |        |
| ✦                                                       | logs               | 03/10/2025 8:34       | Carpeta de archivos |        |
| ✦                                                       | pcaps              | 03/10/2025 8:33       | Carpeta de archivos |        |
| ✦                                                       | recon              | 03/10/2025 8:33       | Carpeta de archivos |        |
| ✦                                                       | report             | 03/10/2025 8:34       | Carpeta de archivos |        |

Enlace al documento word del grupo completo:

[https://docs.google.com/document/d/1qFGQddWwig2uovvcHGtU73LT-3E\\_eHnqQp2YDceZ2v8/edit?usp=sharing](https://docs.google.com/document/d/1qFGQddWwig2uovvcHGtU73LT-3E_eHnqQp2YDceZ2v8/edit?usp=sharing)

**FASE 1 — Reconocimiento y mapeo de red**

**Objetivo:** identificar hosts activos, puertos y servicios.

**Herramientas y comandos (ejemplos):**



- Descubrimiento de hosts:
  - arp-scan (si está permitido):

```
(legui@legui)-[~]
$ sudo arp-scan -l

[sudo] contraseña para legui:
Interface: eth0, type: EN10MB, MAC: 08:00:27:6d:45:bb, IPv4: 192.168.56.17
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.56.22 08:00:27:86:53:97 (Unknown)
192.168.56.40 08:00:27:75:a0:90 (Unknown)
192.168.56.70 08:00:27:7e:fa:03 (Unknown)
192.168.56.75 08:00:27:5f:6d:09 (Unknown)
192.168.56.80 08:00:27:61:0f:54 (Unknown)
192.168.56.81 08:00:27:45:bc:2b (Unknown)
192.168.56.82 08:00:27:59:48:c3 (Unknown)

7 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.872 seconds (136.75 hosts/sec). 7 responded
```

- netdiscover:

```
(legui@legui)-[~]
$ sudo netdiscover -r 192.168.56.0/24
```

```
Currently scanning: Finished! | Screen View: Unique Hosts
44 Captured ARP Req/Rep packets, from 12 hosts. Total size: 2640
```

| IP            | At MAC Address    | Count | Len | MAC Vendor / Hostname  |
|---------------|-------------------|-------|-----|------------------------|
| 192.168.56.22 | 08:00:27:86:53:97 | 3     | 180 | PCS Systemtechnik GmbH |
| 192.168.56.80 | 08:00:27:61:0f:54 | 7     | 420 | PCS Systemtechnik GmbH |
| 192.168.0.67  | 18:60:24:e9:3a:50 | 5     | 300 | Hewlett Packard        |
| 192.168.56.82 | 08:00:27:59:48:c3 | 4     | 240 | PCS Systemtechnik GmbH |
| 192.168.0.65  | e8:39:35:57:9e:e9 | 4     | 240 | Hewlett Packard        |
| 192.168.0.115 | 18:60:24:e9:3b:2a | 3     | 180 | Hewlett Packard        |
| 192.168.0.123 | 34:17:eb:c4:6c:81 | 6     | 360 | Dell Inc.              |
| 192.168.56.40 | 08:00:27:75:a0:90 | 1     | 60  | PCS Systemtechnik GmbH |
| 192.168.56.70 | 08:00:27:7e:fa:03 | 1     | 60  | PCS Systemtechnik GmbH |
| 192.168.56.75 | 08:00:27:5f:6d:09 | 1     | 60  | PCS Systemtechnik GmbH |
| 192.168.56.81 | 08:00:27:45:bc:2b | 1     | 60  | PCS Systemtechnik GmbH |
| 192.168.0.90  | a4:97:33:4e:9f:c4 | 8     | 480 | ASKEY COMPUTER CORP    |

- Escaneo de puertos con nmap:
  - Scan rápido de puertos y servicios:

```
(legui@legui)-[~]
$ nmap -sS -sV -O -Pn 192.168.56.0/24

Starting Nmap 7.95 (https://nmap.org) at 2025-10-03 13:51 CEST
Nmap scan report for 192.168.56.22
Host is up (0.0012s latency).
All 1000 scanned ports on 192.168.56.22 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:86:53:97 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

Nmap scan report for 192.168.56.40
Host is up (0.0014s latency).
All 1000 scanned ports on 192.168.56.40 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:75:A0:90 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop
```

- Escaneo de puertos específicos:

```
(legui@legui)-[~]
$ nmap -p 22,80,443,3128 192.168.56.40

Starting Nmap 7.95 (https://nmap.org) at 2025-10-03 13:58 CEST
Nmap scan report for 192.168.56.40
Host is up (0.0015s latency).

PORT STATE SERVICE
22/tcp closed ssh
80/tcp closed http
443/tcp closed https
3128/tcp closed squid-http
MAC Address: 08:00:27:75:A0:90 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 13.11 seconds
```

- Resultado esperado: lista de hosts, puertos abiertos, servicios y versiones.

#### Tareas del equipo:

- Recon/Red hace mapeo y sube hosts.csv con IP, hostname, puertos y servicios detectados.
- Indicar servicios críticos (SSH, HTTP, RDP, DB, proxy).

## FASE 2 — Análisis de tráfico y paquetes

**Objetivo:** capturar tráfico relevante y analizar patrones/anomalías.

### Herramientas y comandos:

- Captura de paquetes:
  - tcpdump para capturar en interfaz:

```
(legui@legui)-[~]
$ sudo tcpdump -i eth0 -w /home/legui/pcaps/scan_capture.pcap

tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C1193 packets captured
1194 packets received by filter
0 packets dropped by kernel
```

- Wireshark

| No. | Time        | Source                 | Destination   | Protocol | Length | Info                                           |
|-----|-------------|------------------------|---------------|----------|--------|------------------------------------------------|
| 20  | 3.329733858 | Dell_c4:6c:81          | Broadcast     | ARP      | 60     | Who has 192.168.0.203? Tell 192.168.0.123      |
| 21  | 3.789339531 | fe80::aed:edff:feb5... | ff02::1:2     | DHCPv6   | 122    | Solicit XID: 0x64d73d CID: 00010001ffffbcb008e |
| 22  | 3.890712678 | HewlettPacka_e9:3b:... | Broadcast     | ARP      | 60     | Who has 192.168.1.114? Tell 192.168.0.115      |
| 23  | 4.002216146 | MicroStarINT_44:e2:... | Broadcast     | ARP      | 60     | Who has 192.168.0.112? Tell 192.168.0.24       |
| 24  | 4.328137587 | Dell_c4:6c:81          | Broadcast     | ARP      | 60     | Who has 192.168.0.203? Tell 192.168.0.123      |
| 25  | 4.404512044 | 192.168.56.70          | 192.168.56.17 | TCP      | 74     | 46546 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=146 |
| 26  | 4.404677404 | 192.168.56.17          | 192.168.56.70 | TCP      | 54     | 80 → 46546 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0  |
| 27  | 4.408735974 | MicroStarINT_87:2a:... | Broadcast     | ARP      | 60     | Who has 192.168.1.104? Tell 192.168.0.36       |

- Monitoreo en vivo:
  - iftop (consumo por conx):

```
(legui@legui)-[~]
$ sudo iftop -i eth0
```

```
interface: eth0
IP address is: 192.168.56.17
IPv6 address is: fd17:625c:f037:2:84fa:8904:7505:6170
MAC address is: 08:00:27:6d:45:bb
```

| legui@legui: ~ |          |        |       |       |        |               |        |        |        |
|----------------|----------|--------|-------|-------|--------|---------------|--------|--------|--------|
| Session        | Acciones | Editar | Vista | Ayuda |        |               |        |        |        |
|                |          |        |       |       | 12,5Kb | 25,0Kb        | 37,5Kb | 50,0Kb | 62,5Kb |
| 192.168.1.255  |          |        |       |       | ⇒      | 192.168.0.43  |        | 0b     | 0b     |
|                |          |        |       |       | ≤      |               |        | 0b     | 125b   |
| 192.168.1.255  |          |        |       |       | ⇒      | 192.168.0.58  |        | 0b     | 0b     |
|                |          |        |       |       | ≤      |               |        | 0b     | 26b    |
| 192.168.248.2  |          |        |       |       | ⇒      | 224.0.0.1     |        | 0b     | 26b    |
|                |          |        |       |       | ≤      |               |        | 0b     | 0b     |
| 192.168.1.255  |          |        |       |       | ⇒      | 192.168.0.142 |        | 0b     | 0b     |
|                |          |        |       |       | ≤      |               |        | 0b     | 0b     |
|                |          |        |       |       |        |               |        | 0b     | 57b    |
|                |          |        |       |       |        |               |        |        |        |
| TX:            | cum:     | 0B     | peak: | 0b    | rates: | 0b            | 0b     | 0b     |        |
| RX:            |          | 591B   |       | 912b  |        | 0b            | 176b   | 148b   |        |
| TOTAL:         |          | 591B   |       | 912b  |        | 0b            | 176b   | 148b   |        |

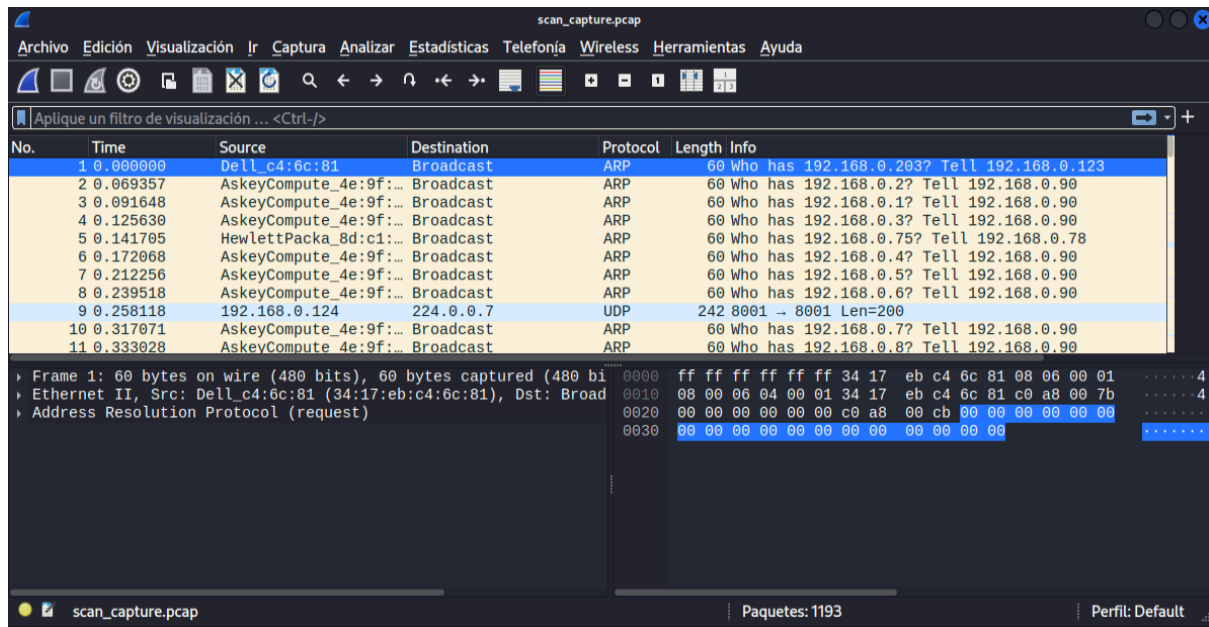
- nethogs (por proceso)

```
(legui@legui)-[~]
$ sudo nethogs
```

NetHogs version 0.8.8

| PID   | USER  | PROGRAM                          | DEV  | SENT  | RECEIVED     |
|-------|-------|----------------------------------|------|-------|--------------|
| 14469 | legui | /usr/lib/firefox-esr/firefox-esr | eth0 | 5.318 | 106.969 kB/s |
| 14701 | legui | /usr/lib/firefox-esr/firefox-esr | eth0 | 0.075 | 0.071 kB/s   |
| ?     | root  | unknown TCP                      |      | 0.000 | 0.000 kB/s   |
| TOTAL |       |                                  |      | 5.393 | 107.039 kB/s |

- Análisis de .pcap:
  - Abrir pcap con **Wireshark** (GUI) o tshark:



```
(legui@legui)-[~]
$ tshark -r /home/legui/pcaps/scan_capture.pcap -q -z io,phs
```

---

```
Protocol Hierarchy Statistics
Filter:
```

|          |                          |
|----------|--------------------------|
| eth      | frames:1193 bytes:106273 |
| arp      | frames:906 bytes:54342   |
| ip       | frames:214 bytes:34316   |
| udp      | frames:204 bytes:33716   |
| data     | frames:140 bytes:17100   |
| ssdp     | frames:15 bytes:2241     |
| xml      | frames:14 bytes:9324     |
| mdns     | frames:16 bytes:2708     |
| llmnr    | frames:2 bytes:132       |
| nbdgm    | frames:5 bytes:1107      |
| smb      | frames:5 bytes:1107      |
| mailslot | frames:5 bytes:1107      |
| browser  | frames:5 bytes:1107      |
| nbns     | frames:12 bytes:1104     |
| igmp     | frames:10 bytes:600      |
| ipv6     | frames:69 bytes:16829    |
| udp      | frames:46 bytes:14923    |
| dhcpcv6  | frames:5 bytes:610       |
| ssdp     | frames:9 bytes:1509      |
| xml      | frames:14 bytes:9604     |
| mdns     | frames:16 bytes:3028     |
| llmnr    | frames:2 bytes:172       |
| icmpv6   | frames:23 bytes:1906     |
| lldp     | frames:4 bytes:786       |

---

- Extracción de flujos/estadísticas con pyshark / scripts Python (opcional).

### Tareas del equipo:

- Capturar tráfico durante un periodo de actividad (p. ej. navegación desde cliente a través de proxy).
- Extraer 3 flujos relevantes y explicar (IP origen/destino, puertos, bytes, flags).
- Detectar posibles anomalías (tráfico inesperado, puertos inusuales).

## FASE 3 — Enumeración de servicios y pruebas seguras

**Objetivo:** verificar configuraciones visibles (vulnerabilidades de servicio *sin explotar*).

### Herramientas:

- nmap scripts (--script), ss, curl, nikto (solo pruebas no destructivas).

- Ejemplos:

```
(legui@legui)-[~]
$ nmap -sV --script=vuln 192.168.56.40
Starting Nmap 7.95 (https://nmap.org) at 2025-10-07 09:51 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.40
Host is up (0.00063s latency).
All 1000 scanned ports on 192.168.56.40 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:75:A0:90 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.79 seconds
```

```
(legui@legui)-[~]
$ curl -I http://192.168.56.40:80

HTTP/1.1 200 OK
Date: Tue, 07 Oct 2025 08:00:40 GMT
Server: Apache/2.4.65 (Debian)
Last-Modified: Thu, 02 Oct 2025 12:22:07 GMT
ETag: "29cf-6402c089f0d09"
Accept-Ranges: bytes
Content-Length: 10703
Vary: Accept-Encoding
Content-Type: text/html
```

```
(legui@legui)-[~]
$ nikto -host http://192.168.56.40:80

- Nikto v2.5.0
+-----+
+ Target IP: 192.168.56.40
+ Target Hostname: 192.168.56.40
+ Target Port: 80
+ Start Time: 2025-10-07 10:01:36 (GMT2)
+-----+
+ Server: Apache/2.4.65 (Debian)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ /: Server may leak inodes via ETags, header found with file /, inode: 29cf, size: 6402c089f0d09, mtime: gzip. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418
+ OPTIONS: Allowed HTTP Methods: HEAD, GET, POST, OPTIONS .
+ 8103 requests: 0 error(s) and 4 item(s) reported on remote host
+ End Time: 2025-10-07 10:02:29 (GMT2) (53 seconds)
+-----+
+ 1 host(s) tested

Portions of the server's headers (Apache/2.4.65) are not in the Nikto 2.5.0 database or are newer than the known string. Would you like to submit this information (*no server specific data*) to CIRT.net for a Nikto update (or you may email to sullo@cirt.net) (y/n)? y

+ ERROR: ->
+ ERROR: Update failed, please notify sullo@cirt.net of the previous line.
```



**Nota:** no realizar explotación sin autorización.

**Tareas:**

- Comprobar encabezados HTTP, existencia de directorios sensibles, versiones de servicios.
- Anotar hallazgos: servicios sin TLS, versiones obsoletas, páginas de administración accesibles, etc.

## FASE 4 — Análisis de logs y forense básico

**Objetivo:** revisar logs del sistema/servicios para identificar actividad sospechosa.

**Ubicaciones y herramientas:**

- Logs de sistema: /var/log/syslog, /var/log/auth.log.

```
(legui@legui)-[~]
$ sudo journalctl
[sudo] contraseña para legui:
oct 02 14:16:55 legui kernel: Linux version 6.12.38+kali-amd64 (devel@kali.org) (x86_64-linux>
oct 02 14:16:55 legui kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.38+kali-amd64 root=>
oct 02 14:16:55 legui kernel: BIOS-provided physical RAM map:
oct 02 14:16:55 legui kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
```

Solo puedo ver los logs de esta forma.

- Logs de aplicaciones:
  - Squid /var/log/squid/access.log
  - web /var/log/apache2/access.log.
- Herramientas: grep, ausearch(auditd), journalctl, ELK/Graylog (si disponible).
- Comandos útiles:
  - sudo tail -n 200 /var/log/auth.log

```
(legui@legui)-[~]
$ sudo journalctl -n 200
oct 07 09:57:09 legui systemd-sysv-generator[37337]: ! This compatibility logic is deprecated
oct 07 09:57:09 legui systemd[1]: Reloading finished in 291 ms.
oct 07 09:57:09 legui systemd[1]: Reload requested from client PID 37212 ('systemctl') (unit
oct 07 09:57:09 legui systemd[1]: Reloading ...
oct 07 09:57:09 legui systemd-sslh-generator: Configuration directory '/etc/sslh/' does not e
oct 07 09:57:09 legui systemd-sysv-generator[37408]: SysV service '/etc/init.d/ptunnel' lacks
oct 07 09:57:09 legui systemd-sysv-generator[37408]: Please update package to include a nativ
oct 07 09:57:09 legui systemd-sysv-generator[37408]: ! This compatibility logic is deprecated
oct 07 09:57:09 legui systemd-sysv-generator[37408]: SysV service '/etc/init.d/inetsim' lacks
```

Solo puedo ver los logs de esta forma.

- `sudo grep "Failed password" /var/log/auth.log | wc -l`

```
(legui@legui)-[~]
$ sudo journalctl _COMM=sshd | grep "Failed password" | wc -l
0
```

Solo puedo ver los logs de esta forma.

```
(legui@legui)-[~]
$ sudo journalctl -u ssh
-- No entries --
```

#### Tareas:

- Identificar intentos fallidos de login, accesos a URLs no autorizadas, timestamps correlacionados con capturas pcap.
- Generar un timeline básico (CSV) con evento, hora, IP origen, descripción.

---

## FASE 5 — Gestión de usuarios y permisos

**Objetivo:** demostrar creación de cuentas, grupos y buenas prácticas y auditar permisos.

#### Comandos (ejemplos):

- Crear usuario y grupo:

```
(legui@legui)-[~]
$ sudo adduser alumno1
sudo groupadd secops

Nueva contraseña:
Vuelva a escribir la nueva contraseña:
passwd: contraseña actualizada correctamente
Cambiando la información de usuario para alumno1
Introduzca el nuevo valor, o pulse INTRO para usar el valor predeterminado
Nombre completo []:
Número de habitación []:
Teléfono del trabajo []:
Teléfono de casa []:
Otro []:
Is the information correct? [Y/n] y

(legui@legui)-[~]
$ sudo groupadd secops

groupadd: el grupo «secops» ya existe
```



```
(legui@legui)-[~]
$ sudo usermod -aG secops alumno1
```

- Asignar permisos a directorio:

```
(legui@legui)-[~]
$ sudo chown root:secops /srv/seguro

(legui@legui)-[~]
$ sudo chmod 750 /srv/seguro
```

- Revisar sudoers (usar visudo):

```
(legui@legui)-[~]
$ sudo visudo
```

```
GNU nano 8.6 /etc/sudoers.tmp
This file MUST be edited with the 'visudo' command as root.

Please consider adding local content in /etc/sudoers.d/ instead of
directly modifying this file.

See the man page for details on how to write a sudoers file.

Defaults env_reset
Defaults mail_badpass
Defaults secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"

This fixes CVE-2005-4890 and possibly breaks some versions of kdesu
(#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults use_pty

This preserves proxy settings from user environments of root
equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

This allows running arbitrary commands, but so does ALL, and it means
different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

While you shouldn't normally run git as root, you need to with etckeeper
```

### Tareas:

- Crear 1 usuario por sub-equipo, asignar grupo y dar acceso restringido a un directorio.
- Documentar comandos y verificar con id usuario y ls -ld.

## FASE 6 — Endurecimiento y medidas de protección (firewall, IDS, mitigaciones)

**Objetivo:** aplicar controles para reducir la superficie de ataque.

**Ejemplos de acciones y comandos:**

### Firewall con nftables (resumen)

- Tabla y cadena básica:

```
(legui@legui)-[~]
$ sudo nft add table inet filtro
```

```
(legui@legui)-[~]
$ sudo nft add chain inet filtro input '{ type filter hook input priority 0; policy drop; }'
```

```
(legui@legui)-[~]
$ sudo nft add rule inet filtro input iif lo accept
sudo nft add rule inet filtro input tcp dport '{22,80,443,3128}' accept
sudo nft add rule inet filtro input ct state '{established,related}' accept
```

- Guardar reglas:
  - sudo nft list ruleset > /etc/nftables.conf

```
(legui@legui)-[~]
$ sudo nft list ruleset | sudo tee /etc/nftables.conf > /dev/null
```

Solo me deja de esta forma

### Firewall con ufw (sencillo)

```
(legui@legui)-[~]
$ sudo ufw allow 22/tcp
sudo ufw allow 3128/tcp
sudo ufw enable

Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Firewall is active and enabled on system startup
```

## HIPS / Hardening

- Habilitar fail2ban para SSH:

```
(legui@legui)-[~]
$ sudo apt install fail2ban
sudo systemctl enable --now fail2ban

Installing:
 fail2ban

Installing dependencies:
 python3-systemd

Paquetes sugeridos:
 mailx system-log-daemon monit

Summary:
 Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 669
 Download size: 507 kB
 Space needed: 2.461 kB / 8.019 MB available
```

- Auditar con lynis:

```
(legui@legui)-[~]
$ sudo apt install lynis
sudo lynis audit system

Installing:
 lynis

Installing dependencies:
 menu

Paquetes sugeridos:
 apt-listbugs debsecan debsums tripwire samhain aide menu-l10n

Summary:
 Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 669
 Download size: 629 kB
 Space needed: 3.282 kB / 8.013 MB available

Continue? [S/n] s
Des:1 http://mirror.es.cdn-perfprod.com/kali kali-rolling/main amd64 lynis all 3.1.4-1 [273 kB]
Des:2 http://kali.download/kali kali-rolling/main amd64 menu amd64 2.1.51 [355 kB]
Descargados 629 kB en 54s (11,7 kB/s)
Seleccionando el paquete lynis previamente no seleccionado.
(Leyendo la base de datos ... 419747 ficheros o directorios instalados actualmente.)
```

**Tareas:**

- Aplicar reglas mínimas de firewall que permitan solo lo necesario.
- Activar fail2ban y demostrar que bloquea after X intentos (simulación).
- Documentar configuración y justificar decisiones.

**FASE 7 — Informe final y presentación****Entregables (mínimos):**

1. report/Informe\_EquipoX.pdf con:
  - Alcance y reglas del laboratorio.
  - Resumen de hallazgos (hosts, puertos, servicios).
  - Evidencias (pcap snippets, logs extractados, capturas).
  - Riesgos priorizados y medidas de remediación (con comandos).
  - Cronograma y roles.
2. Carpeta pcaps/, logs/, scripts/.
3. Diapositiva para presentación

**Herramientas del módulo «Seguridad informática» (recordatorio para la práctica)**

- Recon / Scanning: nmap, netdiscover, arp-scan, masscan (con precaución).
- Captura / Análisis de tráfico: tcpdump, tshark, Wireshark, iftop, nethogs.
- Packet crafting / análisis: scapy, pyshark.
- Logs / SIEM básico: journalctl, auditd, rsyslog, ELK (Elasticsearch + Logstash + Kibana) o Splunk (si disponible).
- Forense / integridad: sleuthkit, autopsy, chkrootkit, rkhunter.
- Hardening / auditoría: ufw, nftables, iptables (legacy), fail2ban, lynis.
- Proxy / Web: squid, apache2, nginx, nikto.
- Gestión de usuarios/permisos: adduser, groupadd, usermod, chmod, chown, visudo.
- Scripting y automatización: Bash, Python (pandas, scikit-learn opcional), Streamlit (visual).
- Documentación / versiones: git.

**Ejercicios concretos para entregar (lista rápida)**

1. recon/hosts.csv + nmap output.
2. pcaps/pcap1.pcap con explicación de 3 flujos.
3. logs/logs\_issue.txt con 3 hallazgos importantes extraídos de logs.
4. hardening/ con nftables.conf y pasos ejecutados (comandos).
5. report/Informe\_EquipoX.pdf y 5–8 diapositivas resumen.

Optativo quien quiera realizar una exposición en grupo del trabajo realizado o al menos explicar al resto de compañeros lo que han realizado.